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Lundberg

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(54) **PATENT MAPPING**

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(21) Appl. No.: **19/262,064**

(22) Filed: **Jul. 7, 2025**

Related U.S. Application Data

(63) Continuation of application No. 17/103,870, filed on Nov. 24, 2020, now Pat. No. 12,380,171, which is a continuation of application No. 13/253,936, filed on Oct. 5, 2011, now Pat. No. 10,860,657.

(60) Provisional application No. 61/542,515, filed on Oct. 3, 2011.

Publication Classification

(51) **Int. Cl.**

G06F 16/93 (2019.01)
G06F 16/21 (2019.01)
G06F 16/23 (2019.01)
G06F 16/2457 (2019.01)
G06F 16/248 (2019.01)
G06F 16/25 (2019.01)
G06F 16/334 (2025.01)
G06F 16/338 (2019.01)
G06F 16/35 (2025.01)
G06F 16/355 (2025.01)
G06F 16/36 (2019.01)

G06F 16/38 (2019.01)

G06F 16/903 (2019.01)

G06F 16/951 (2019.01)

G06F 16/9535 (2019.01)

G06Q 10/06 (2023.01)

G06Q 10/0631 (2023.01)

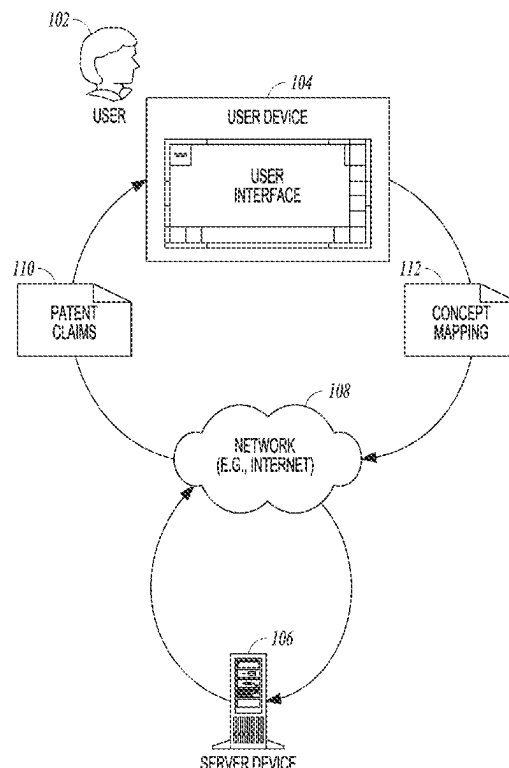
G06Q 50/18 (2012.01)

(52) **U.S. Cl.**

CPC **G06F 16/93** (2019.01); **G06F 16/21** (2019.01); **G06F 16/23** (2019.01); **G06F 16/2358** (2019.01); **G06F 16/24578** (2019.01); **G06F 16/248** (2019.01); **G06F 16/25** (2019.01); **G06F 16/334** (2019.01); **G06F 16/338** (2019.01); **G06F 16/35** (2019.01); **G06F 16/355** (2019.01); **G06F 16/367** (2019.01); **G06F 16/38** (2019.01); **G06F 16/951** (2019.01); **G06F 16/9535** (2019.01); **G06Q 10/06** (2013.01); **G06Q 10/0631** (2013.01); **G06Q 50/184** (2013.01); **G06F 16/211** (2019.01); **G06F 16/90335** (2019.01)

ABSTRACT

System and method permit patent mapping. A method may comprise maintaining a database of patent portfolios and a database of patents with each patent stored in the database of patents associated with one or more patent portfolios stored in the database of patent portfolios. A target subject matter to be mapped is identified and a search query associated with the target subject matter is received. Search results are generated to define a first patent portfolio in the database with the search results including one or more patent claims associated with the search query. The one or more patent claims are mapped to a patent concept.



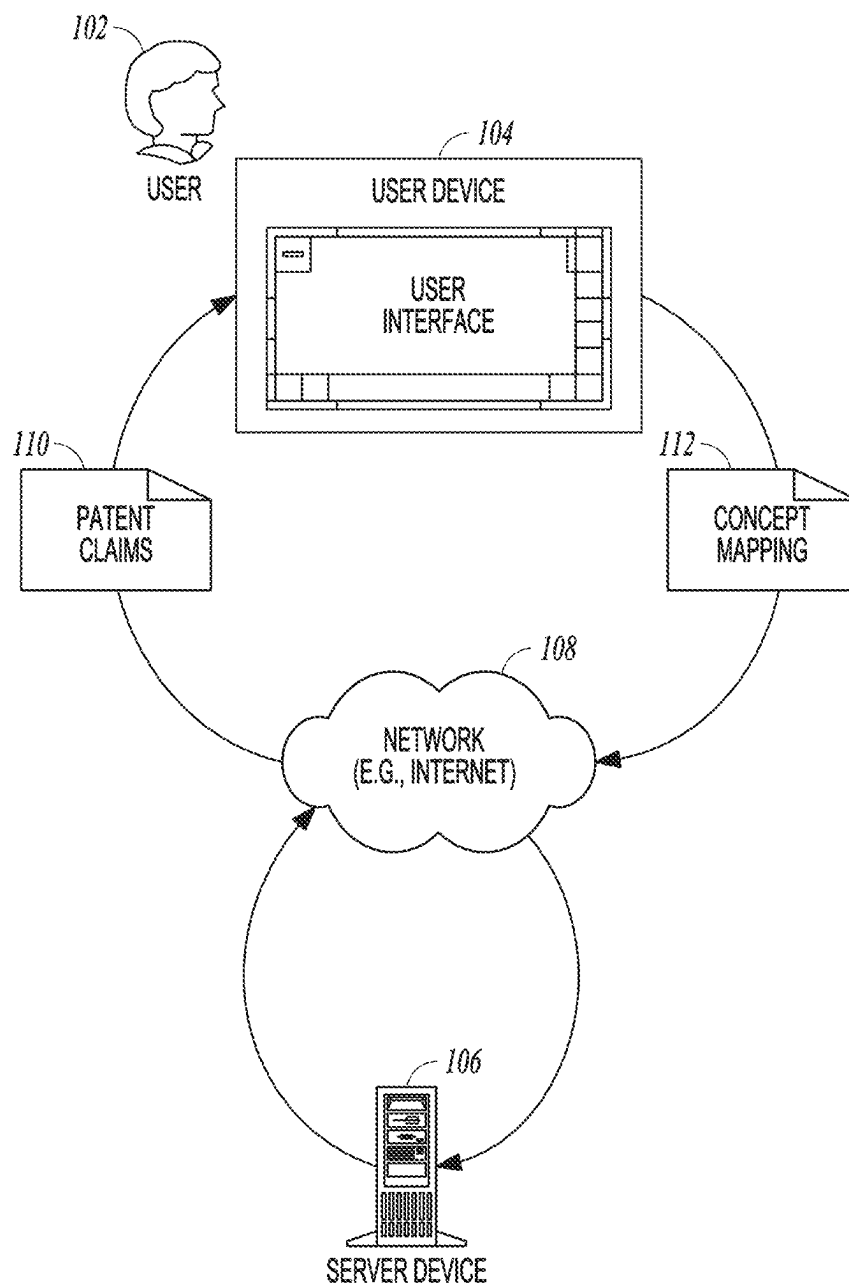


FIG. 1

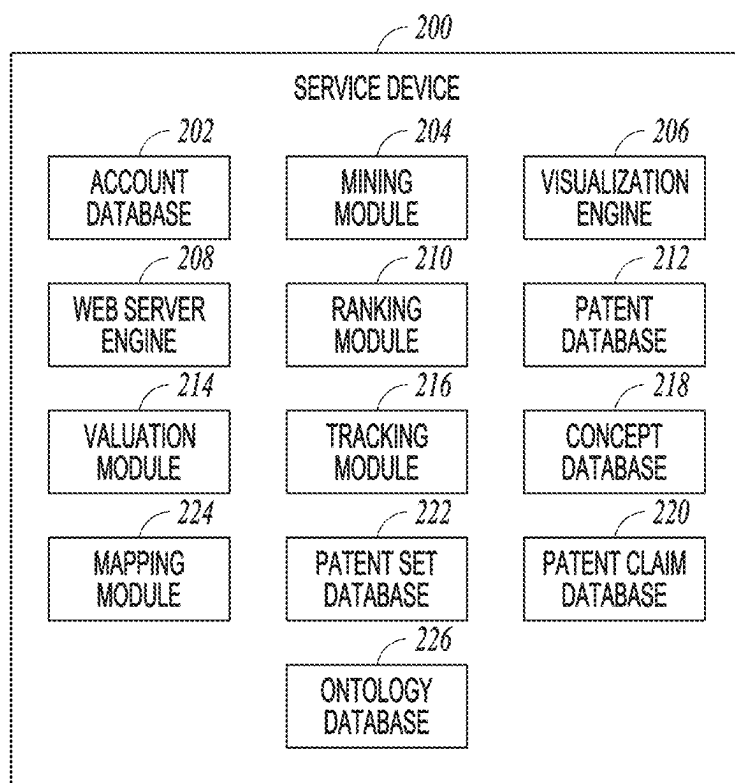


FIG. 2

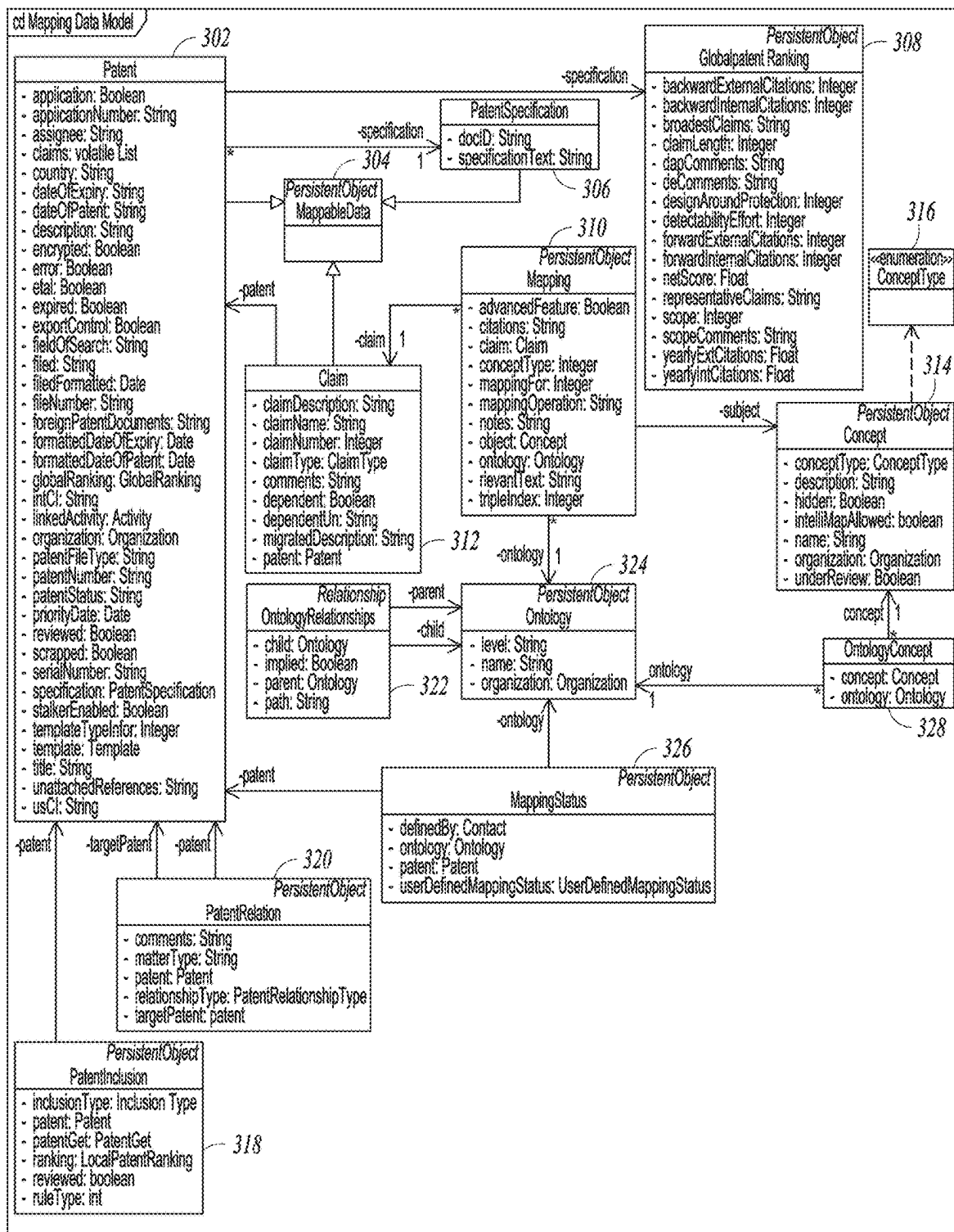


FIG. 3

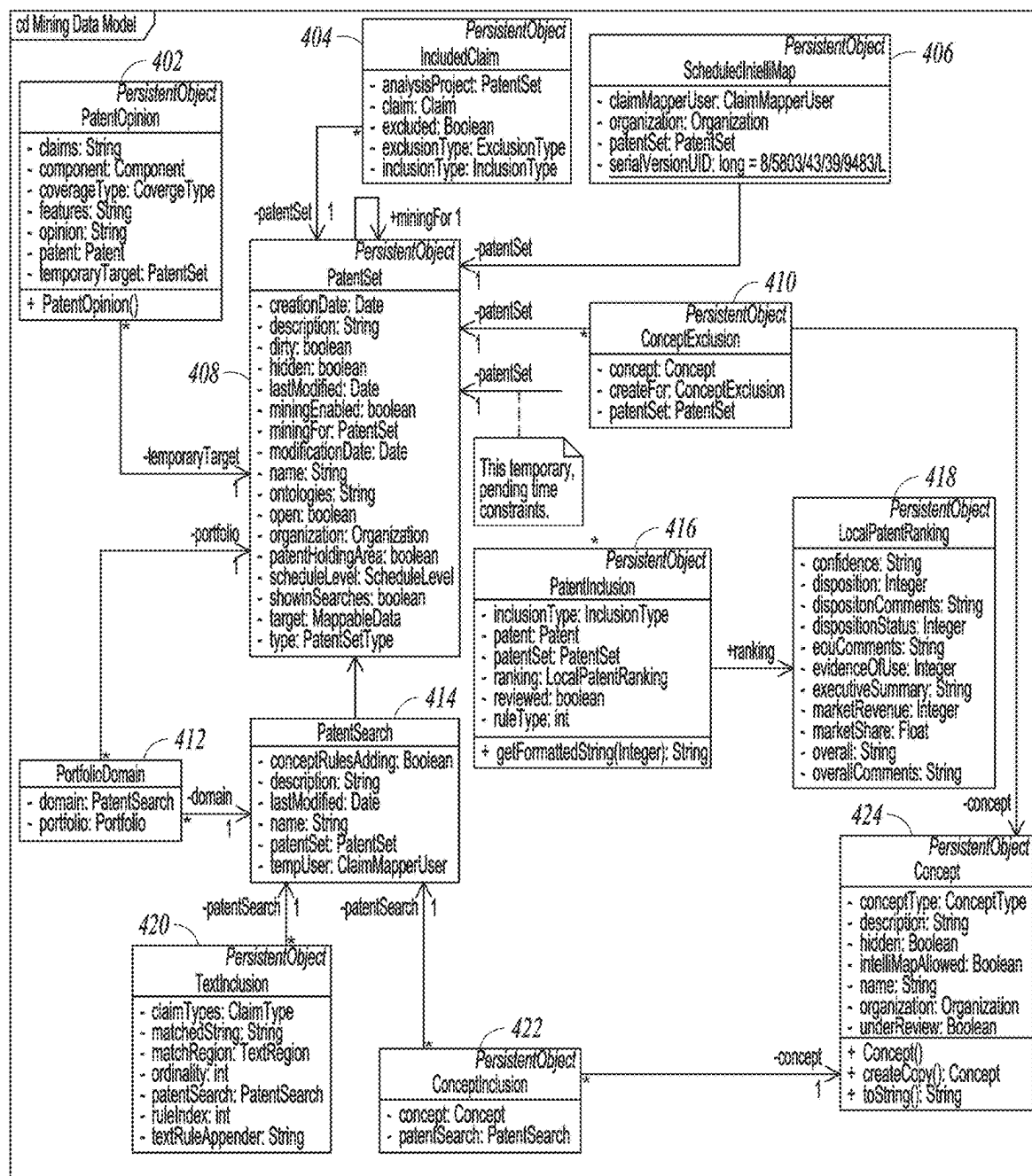


FIG. 4

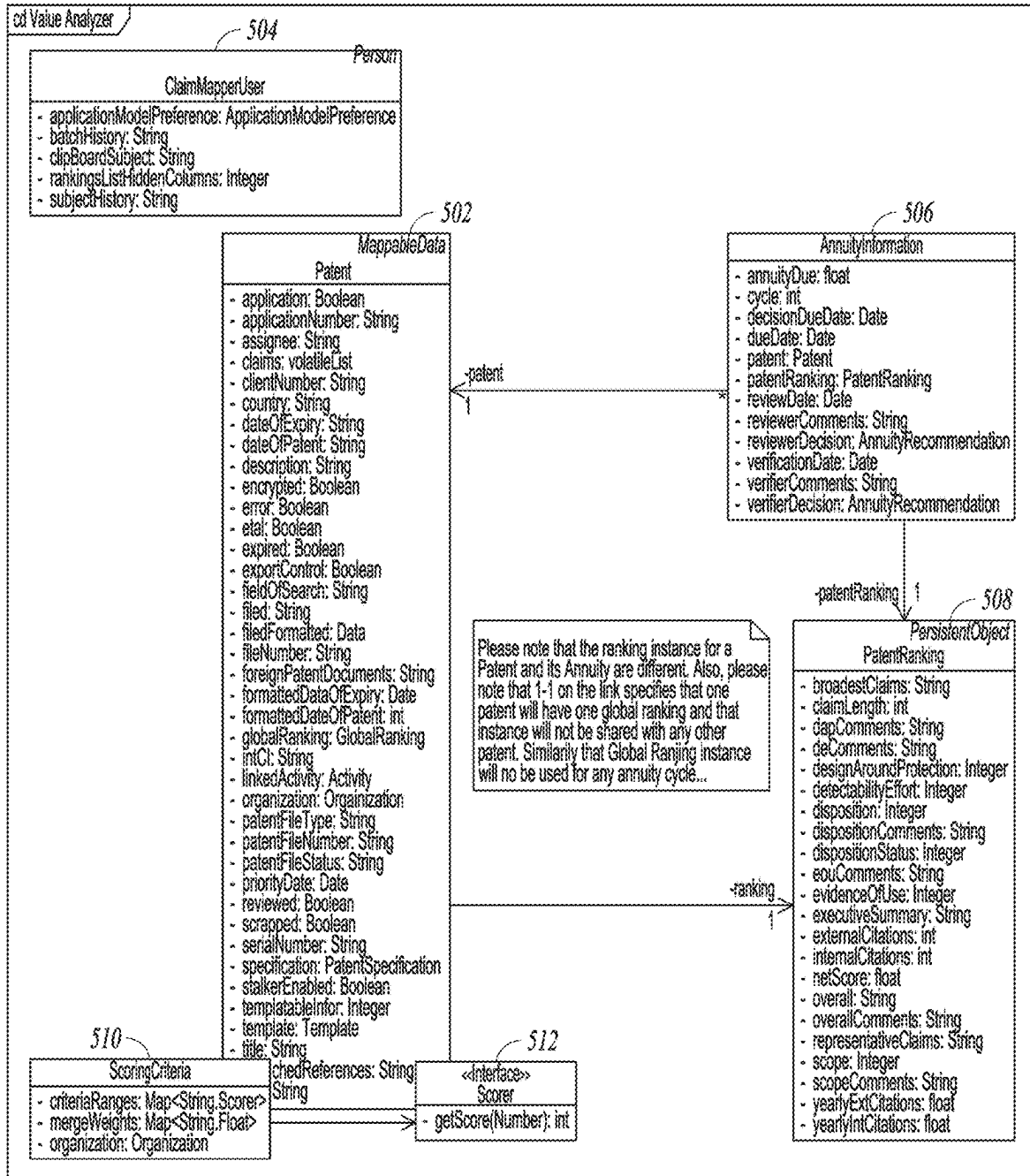


FIG. 5

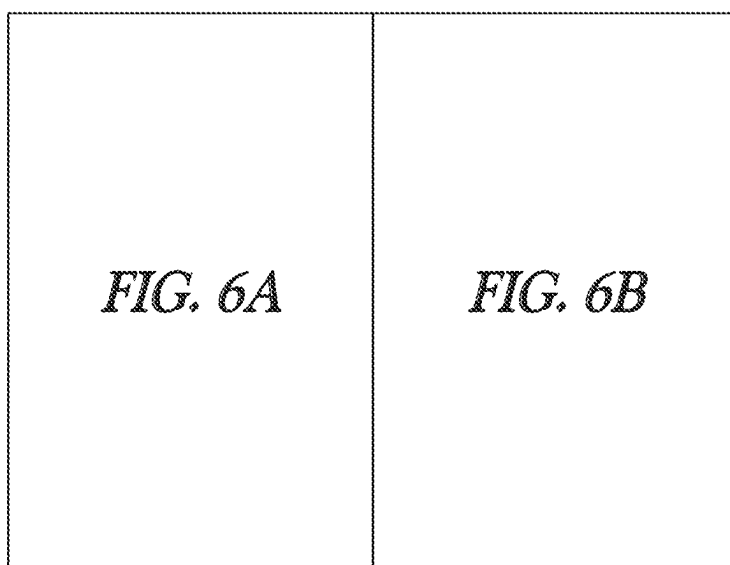


FIG. 6

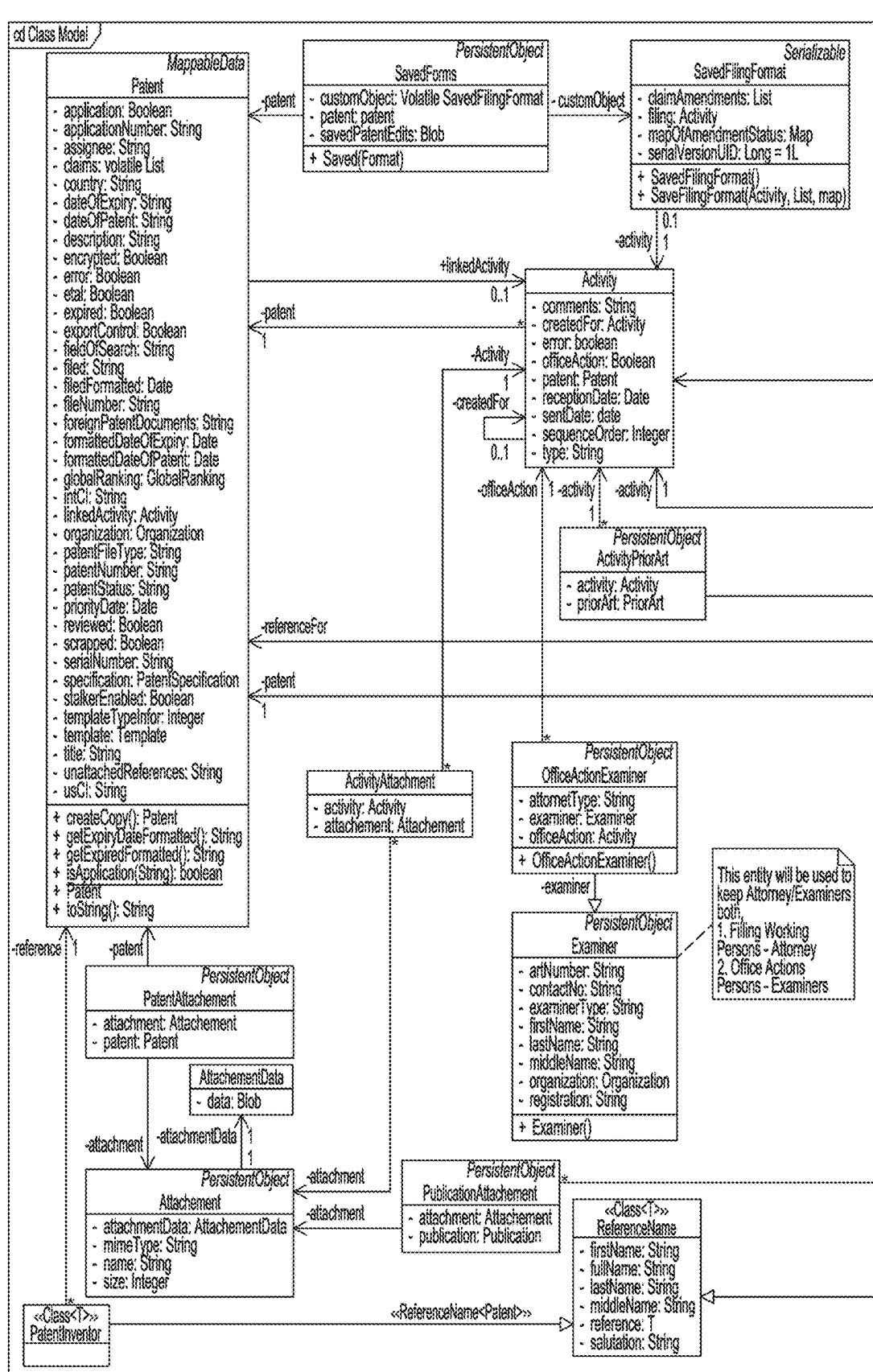


FIG. 6A

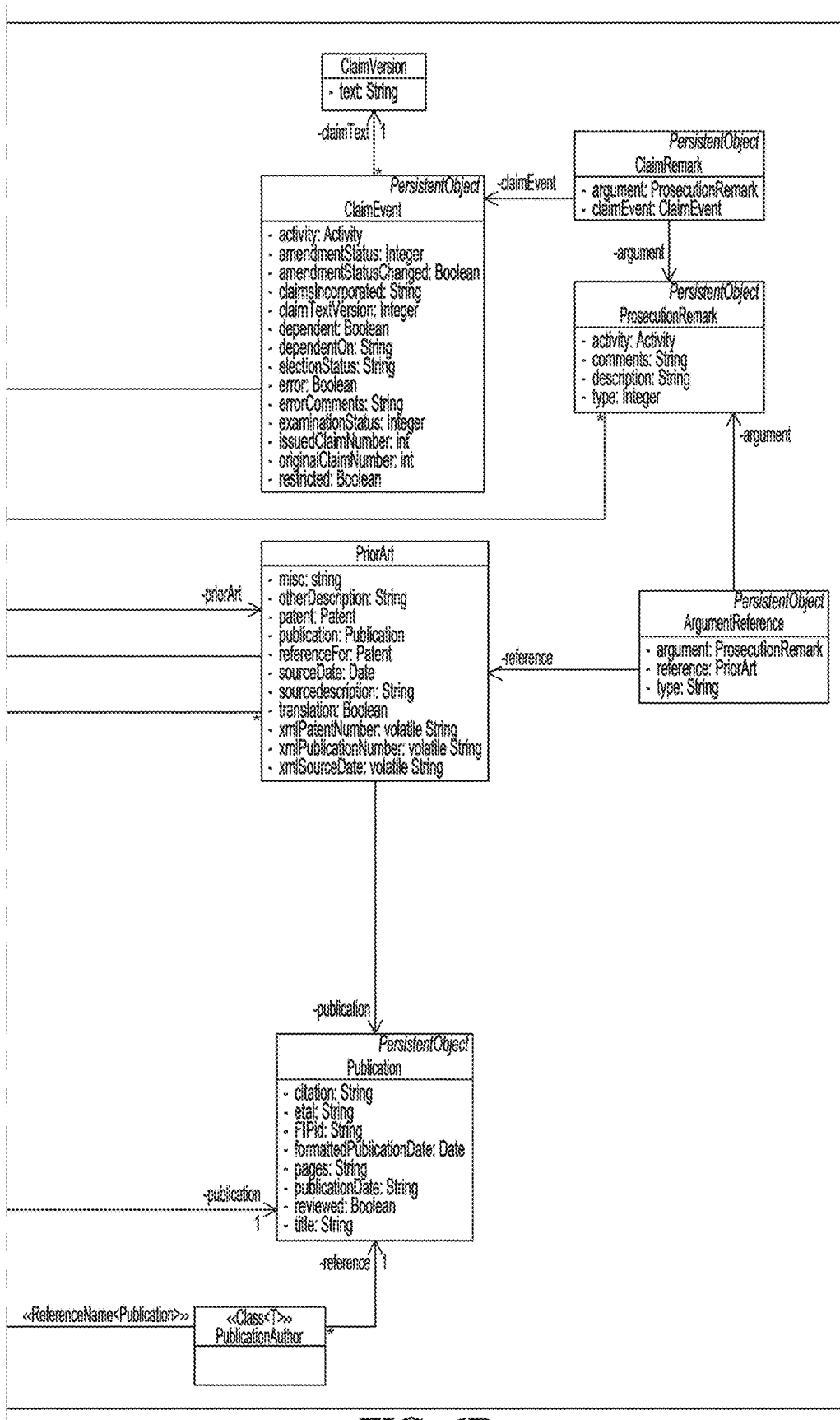


FIG. 6B

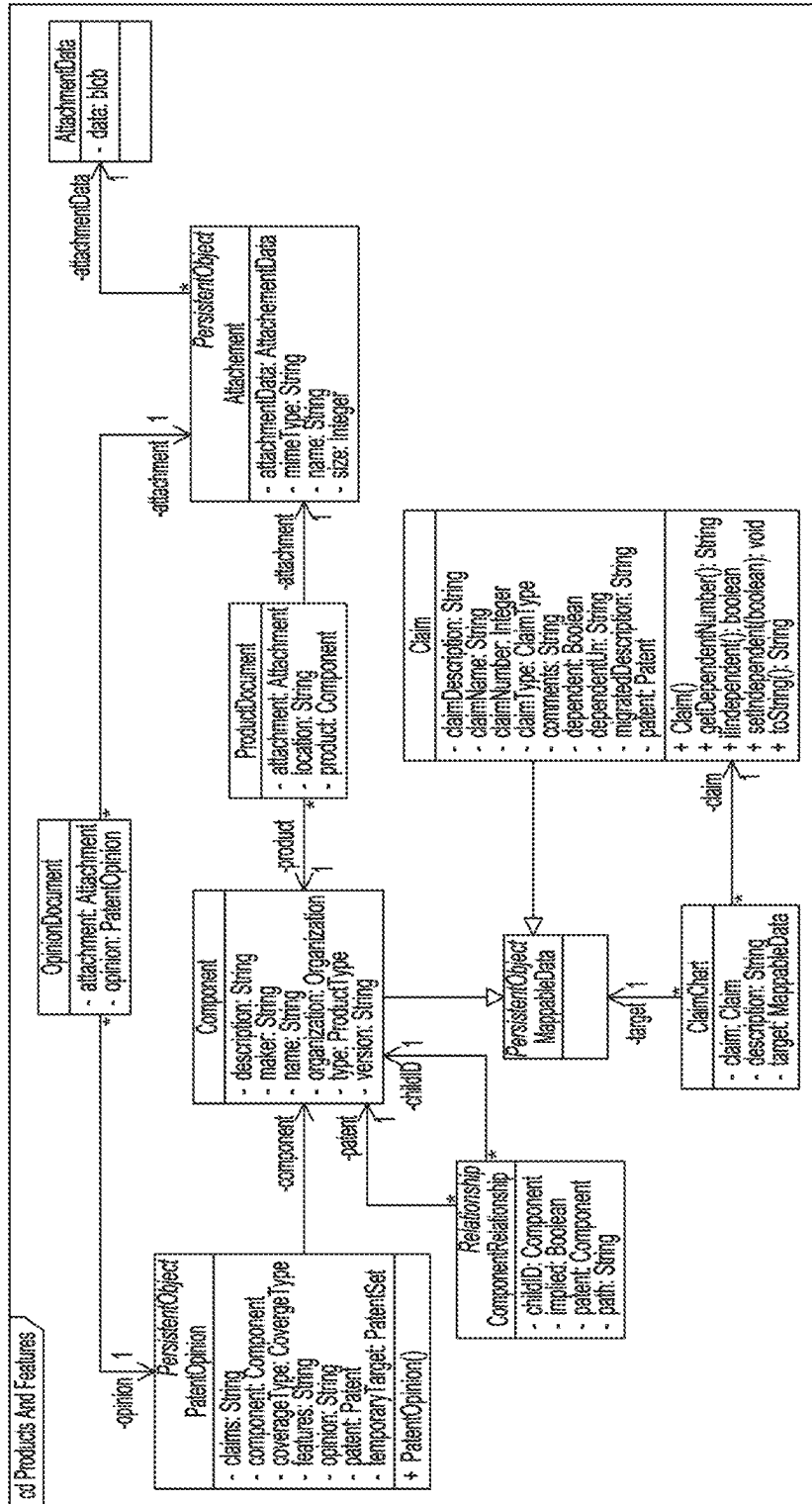


FIG. 7

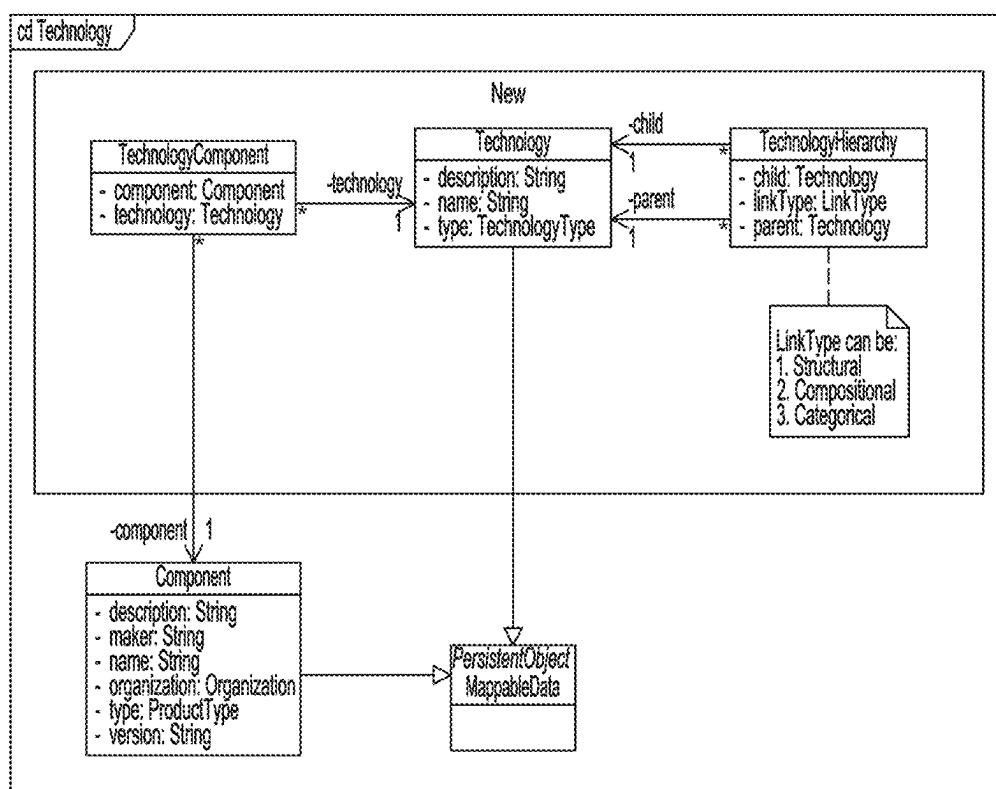


FIG. 8

Matter | Portfolio | Product | Ontology | Report | Search | Contact

Patent#

Portfolio : Omni Map

Portfolio

Edit | List | Quick Rank | Panoramic Claim Map

Status

Open

Title

Perpetual Motion

Created On

Aug 06, 2008

Default Ontology

Perpetual Motion

Last Modified

Aug 06, 2008

Details

Matters

Rankings

Holding Area

Participants

Search

Omni Map

Search Criteria

☒ Independent Claims

☐ Dependent Claims

☐ Both

☒ *

☒ Searched Keywords

☐ Mapped Keywords

☒ Select Tech Category(s)

☐ Tech Category(s) are -

☐ And

☒ Or

☒ Show TC

☒ Select Scope Concept(s)

☐ Scope Concept(s) are -

☒ And

☒ Show SC

Search

Reset Filters

Map TC

Map SC

926

DM Directly Mapped, DNM Does Not Map, U Unresolved, NO No Operation

FIG. 9

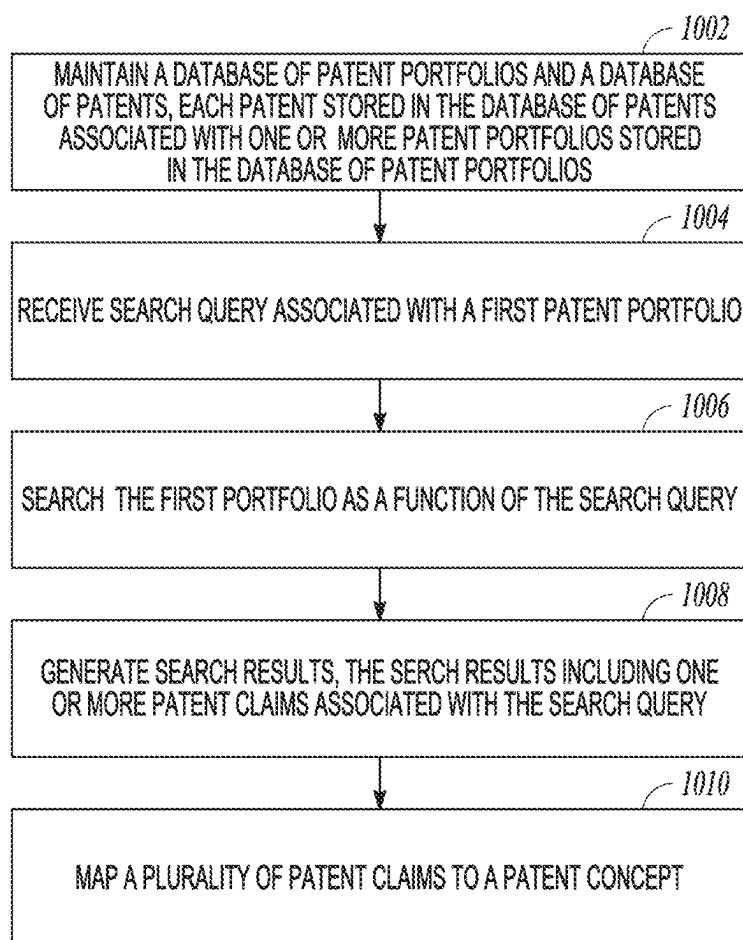


FIG. 10

Ontology Concept(s)

Back to Ontology Page

Search Criteria

Clear Filters

Where

Text

*

Show Inherited

☐

New Portfolio Ontology
openly responses
Perpetual Motion

Go

Search

☐ Technology Category

☐ Keyword

☒ Scope Concept

Go

Show

100

root concept per page.

Go to Page

Go

[First] [Previous] Concept 1 to 2 of 2 [Next] [Last]

ScopeConcept

Bulk Add

Add new

Clipboard Concept

Claim(s) Count

Ontology

Concept	Claim(s) Count	Indirect Claims Count	Hier. Claim(s) Count	Ontology	Actions
fluid with randomly moving particles	3	0	3	Perpetual Motion	
flux modulation controller generating an electromagnetic field or a gravitomagnetic field	3	0	3	Perpetual Motion	

FIG. 11

Search Criteria

<<

Search Claims	☒ Independent Claims ☐ Dependent Claims ☐ Both			Search
Search Expression	☒ razor && handle			Reset Filters
Highlight Keywords	☒ Searched Keywords ☐ Mapped Keywords			
	Select Tech Category(s) Tech Category(s) are - ☐ And ☒ Or ☒ Show TC Clear Tech Category(s)			
Tech Category(s)	Technology Concept Name	Filter		
	Liquid Dispensing Razors	☒ Direct Mapped Claims		
		☐ Direct Mapped or 'Does Not Map'		
	☐ Direct Mapped or 'Unresolved'			
Scope Concept(s)	Select Scope Concept(s)	Scope Concept(s) are - ☒ And ☐ Or	☒ Show SC	

FIG. 12

Scope Concept(s)	Scope Concept Name	Filter
Multiple Blades		☐ Direct Mapped Claims
		☒ Do not include Direct Mapped and 'Does Not Map'
		☐ Direct Mapped or 'Does Not Map'
		☐ Direct Mapped or 'Unresolved'

FIG. 13

1406 Add_New | Cancel | Return_to_Mapping Help

1402 Search:

Where Text

Show Inherited ☐ Selected Ontology Razors - Shaving Search Clear

☐ Intelli-Search

Search Results

Showing 100 Concepts per page. Go to Page Go

[First] [Previous] Concept 1 to 1 of 1 [Next] [Last]

1404 Map

1410	Category/Concept	Description	Ontology	Action
1408	☐ 𐄂 multiple blades	Subject created by Claimbot	Razors - Shaving	1412 Map Modify and Map

FIG. 14

Create/Edit Concept

Save Cancel

Concept Name

Description

Concept Type

Attach To Ontology

FIG. 15

Title							Title #1	Title #2
Patent#							6,697,xxx	7,272,xxx
Filing Date							Oct 01, 1999	Jan 14, 20
Total Claims							58	28
Claim# (Independent)							1 11 30 44 52	1 14 22
Owner							Acme Co.	Dynamite,
Scope Concept (Frequency, Desc.)								
SC #1							1 1 1 1 1 1 1 1	1 1 1 1
SC #2								

FIG. 16

Title				Title 1	Title 2	Title 3
Patent#				6,697,xxx	7,272,xxx	7,327,xxx
Filing Date				Oct 01, 1999	Jan 14, 2004	Mar 02, 2004
Total Claims				58	28	24
Claim# (Independent)				1 11 30 44 52	1 14 22 27	1 11 18 24
Claim Coverage				3 3 3 3 3	1 1 1 1 1	2 2 2 2 2
Product Coverage				Potentially Applies	Doesn't Apply	Not sure
Has Feature?						
Scope Concept (Frequency, Desc.)	Rating (1-3)	Total		2 2 2 2 2	2 2 2 2 2	1 1 1 1 1
SC 1	3	20	38%			
SC 2	2	7	13%			
SC 3	3	7	13%			
SC 4	1	6	12%			
SC 5	3	5	10%			
SC 6	1	4	8%			

FIG. 17

Title			Title 1	Title 2	Title 3
Patent#			6,697,xxx	7,272,xxx	7,327,xxx
Filing Date			Oct 01, 1999	Jan 14, 2004	Mar 02, 2004
Total Claims			58	28	24
Claim# (Independent)			1 11 30 44 52	1 14 22 27	1 11 18 24
Claim Value			1 1 1 1 1	3 3 3	2 2 2 2
Potential Patent Value			Not Important	Essential	Important
Need Feature?					
Scope Concept (Frequency, Desc.)	Rating (1-3)	Total	2 2 2 2 2	2 2 2	1 1 1 1
SC 1	3	20	38%		
SC 2	2	7	13%		
SC 3	3	7	13%		
SC 4	1	6	12%		
SC 5	1	5	10%		
SC 6	3	4	8%		

FIG. 19

Title	Title	Title 1	Title 2	Title 3
Patent#		6,697,xxx	7,272,xxx	7,327,xxx
Filing Date		Oct 01, 1999	Jan 14, 2004	Mar 02, 2004
Total Claims		58	28	24
Claim# (Independent)		1 11 30 44 52	1 14 22 27	1 11 18 24
Claim Status		2 2 2 2 2	3 3 3 3	1 1 1 1 1
Feature rating		Some Overlap	Complete Overlap	No Overlap
Feature Shown?				
Scope Concept (Frequency, Desc.)	Rating (1-3)	Total		
SC 1	3	20	38%	
SC 2	1	7	13%	
SC 3	3	7	13%	
SC 4	1	6	12%	
SC 5	1	5	10%	
SC 6	3	4	8%	

FIG. 20

Claim Concept White Space Analysis		
Used Scope Concepts in Red		
Unused/Unclaimed Scope Concepts in Blue		
Scope Concept (Frequency, Desc.)	Number of Uses	Percent
SC 1	20	38%
SC 2	7	13%
SC 3	7	13%
SC 4	6	12%
SC 5	5	10%
SC 6	not found	0%
SC 7	not found	0%

FIG. 21

Title		Unclaimed Combinations of Scope Concepts from Patent Set												New Combinations Using New Concepts in Combination with Existing Concepts									
		1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10
Proposed Claim# (Independent)																							
Scope Concept (Frequency, Desc.)	Old/New	2	2	2	2	2	1	1	1	2	2	2	2	1	1	1	1	1	1	1	1	1	1
SC 1	old																						
SC 2	old																						
SC 3	old																						
SC 4	old																						
SC 5	old																						
SC 6	old																						
SC 7	old																						
SC 8	old																						
SC 9	old																						
SC 10	old																						
SC 11	old																						
SC 12	old																						
SC 13	old																						
SC 14	new																						
SC 15	new																						
SC 16	new																						
SC 17	new																						
SC 18	new																						

FIG. 22

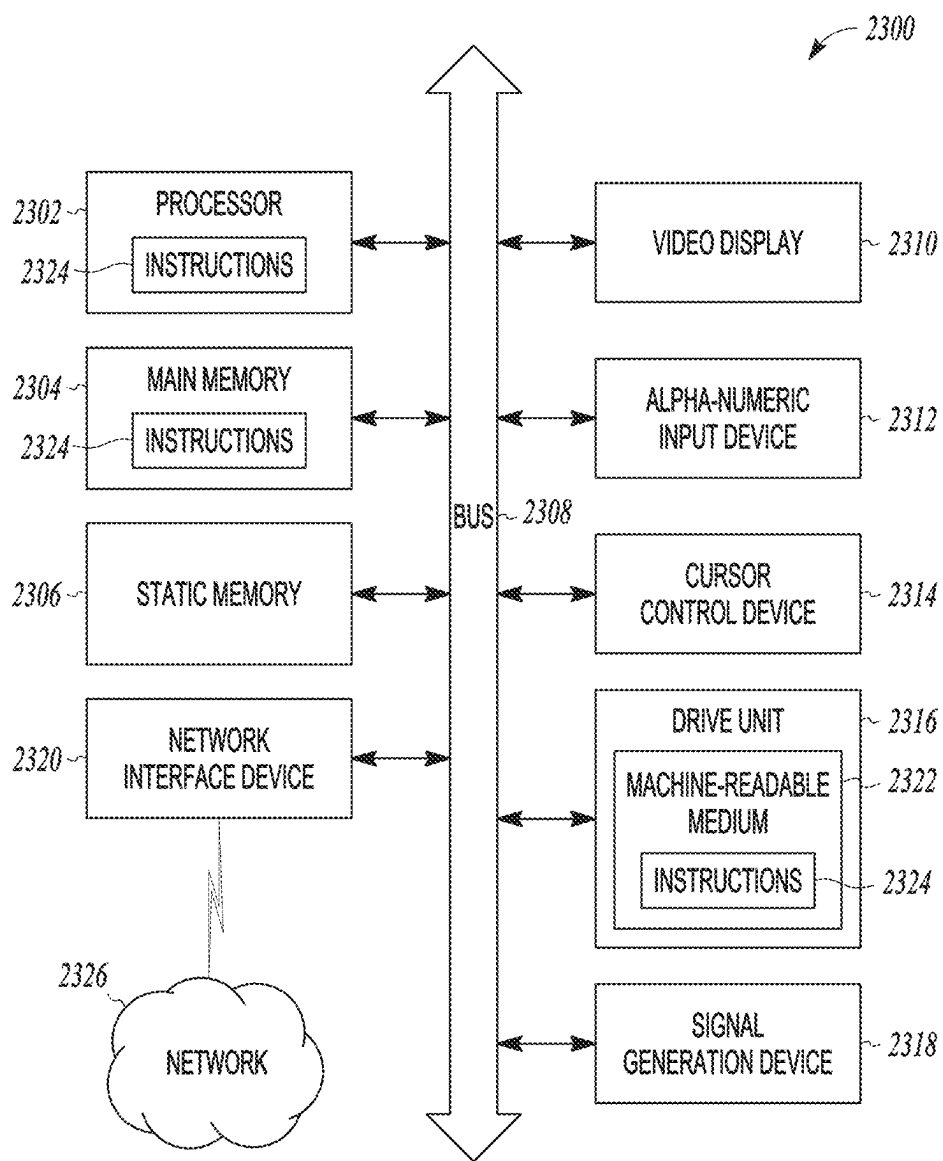


FIG. 23

Matter | Portfolio | Reference | Contact

Patent # ▾

Portfolio

Save Cancel

Title:

Description:

Review Recording

☒ On ☐ Off (Patent mapping confidence will be displayed along with mapping status)

Mapping Status Indicator

☒ On ☐ Off (If enabled, application will automatically set mapping status incomplete/

Type

Analysis Portfolio ▾

Analysis Portfolio

Prior Art Portfolio

240

☐	<u>test</u>	2	June 05, 2011	<u>test</u>
☐	<u>test</u>	1	June 05, 2011	<u>Create/Assign</u>
☐	<u>test</u>	2	June 05, 2011	<u>Create/Assign</u>

FIG. 24

List View

Chart View

Remove

Bulk Update Status

Show Dismissed

Add New

Patent Reference

Go

250

US Patents

☐	Patent/Pub#	Grant/Pub Date	Inventor	Flow level	Source Date	Status	Actions
☐	1000001	Apr 01, 2011	Last1up, First2 M	0	Jun 02, 2011	Unmarked	View Edit

Foreign Patents

☐	Patent/Pub#	Country	Inventor	Flow level	Source Date	Status	Actions
☐	67892462	Thailand	Test1		Jun 13, 2011	Received from PTO	View Edit

Non Patent Reference

☐	Author	Grant/Pub Date	Inventor	Flow level	Source Date	Status	Actions
☐	sss	Jun 30, 2011	ss	0	Jun 02, 2011	Unmarked	View Edit

FIG. 25

Misc Information

Status

Translation

Attachment

Unmarked

Nul

Cite

Unmarked

Received From PTO

Dismissed

260

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Save & New

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FIG. 26

List View

Chart View

Remove

Bulk Update Status

272

270

Show Dismissed

Add New

Patent Reference

Go

US Patents

☐	Patent/Pub#	Grant/Pub Date	Inventor	Flow level	Source Date	Status	Actions
☐	1000001	Apr 01, 2011	Last1up, First2 M	0	Jun 02, 2011	Unmarked	View Edit

Foreign Patents

☐	Patent/Pub#	Country	Inventor	Flow level	Source Date	Status	Actions
☐	67892462	Thailand	Test1		Jun 13, 2011	Received from PTO	View Edit

FIG. 27

List View

Chart View

☐ Remove

Show Dismissed ☐

Add New

US Patents

☐	Patent/Pub#	Grant/Pub Date	Inventor	Flow level	Source Date	Status	Actions
☐	1000001	Apr 01, 2011	Last1up, First2 M	0	Jun 02, 2011	Unmarked	View Edit

Foreign Patents

☐	Patent/Pub#	Country	Inventor	Flow level	Source Date	Status	Actions
☐	67892462	Thailand	Test1		Jun 13, 2011	Received from PTO	View Edit

Non Patent Reference

☐	Author	Grant/Pub Date	Inventor	Flow level	Source Date	Status	Actions
☐	sss	Jun 30, 2011	ss	0	Jun 02, 2011	Unmarked	View Edit

FIG. 28

LUCID PATENT

Matter | Portfolio | Reference | Contact

Claim Mapping | Claim Tracking | Logout | Setup

Patent # ▾

Portfolio: Reference(s)

Portfolio	Panoramic Claim Map	Matter Count	2
Title	Ⓢ Ⓜ (PA) test	Created On	June 02, 2011
Default Ontology	test	Last Modified	Jun 13, 2011

Details | Participants | Matters | References

All References | Patent References | Non-Patent References | Bulk Add | Job Status | Holding List

▷ Search Criteria

Delete

Add Patent Reference | Add Non Patent Reference

☐	Official Number	Unofficial Number	Title	Inventor/Author Name	Issue/Publication Date	Effective Date	Type	Attachment
☐	67,892,462	MAN-669650		Test1 ;			Patent Reference	

FIG. 29

Portfolio: Reference(s)

Portfolio	Panoramic Claim Map	Matter Count	2
Title	test	Created On	June 02, 2011
Default Ontology	test	Last Modified	Jun 13, 2011

Details

Participants

Matters

References

Search

Quick Rank

Ontology

Omni Map

All References

Patent References

Non-Patent References

Bulk Add

Job Status

Holding List

test: Scrape References

Please enter references to be added into Portfolio

Job Name test

FIG. 30

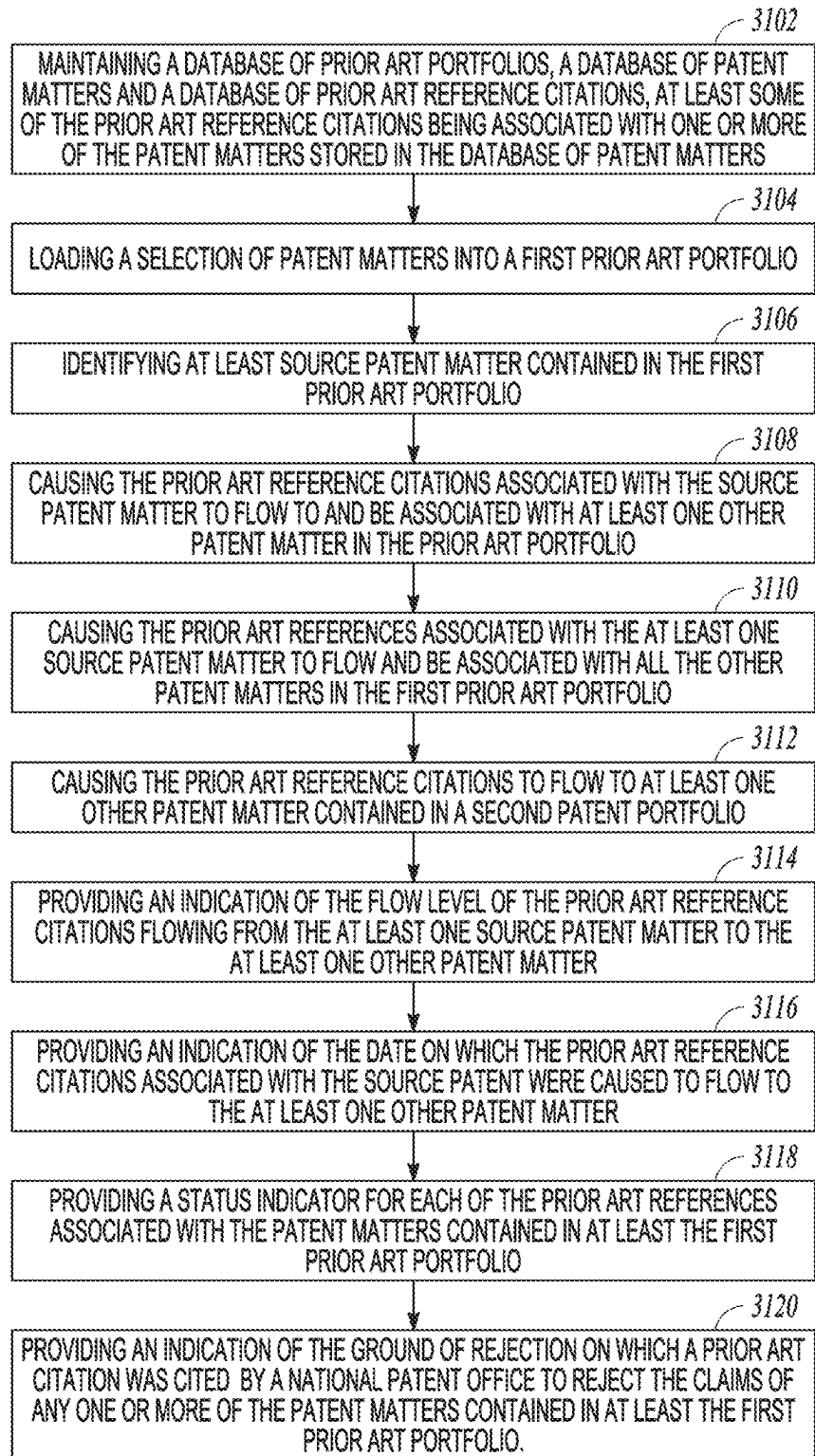


FIG. 31

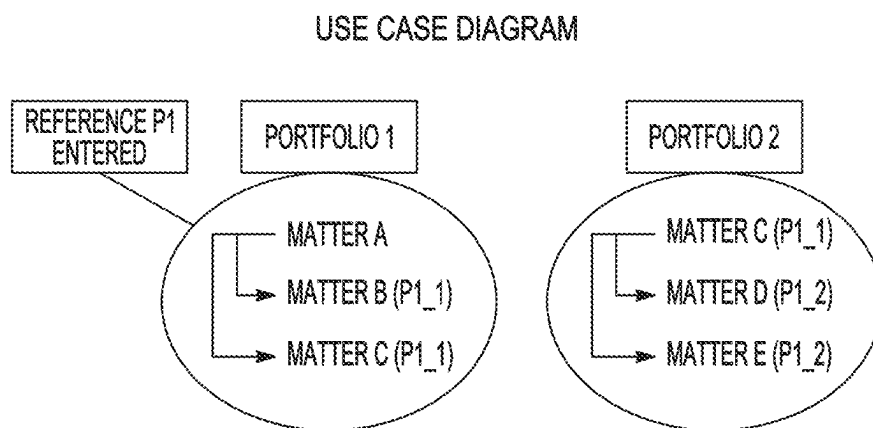


FIG. 32

PATENT MAPPING

RELATED APPLICATIONS

[0001] This application is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/103,870, filed on Nov. 24, 2020, which is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/253,936, filed on Oct. 5, 2011, now U.S. Pat. No. 10,860,657, issued on Dec. 8, 2020, which claims the benefit under 35 U.S.C. § 119 (e) of U.S. Provisional Patent Application Ser. No. 61/542,515 filed Oct. 3, 2011, each of which is incorporated herein by reference in its entirety and made a part hereof.

[0002] This application is related to U.S. patent application Ser. No. 12/605,030, entitled “Patent Mapping” by Steve W. Lundberg and Thomas G. Marlow filed Oct. 23, 2009, which is incorporated herein by reference in its entirety and made a part hereof. U.S. patent application Ser. No. 12/605,030 claims the benefit under 35 U.S.C. 119 (e) of U.S. Provisional Patent Application Ser. No. 61/107,930 filed Oct. 23, 2008, which is also incorporated herein by reference in its entirety and made a part hereof.

[0003] This application is further related to U.S. patent application Ser. No. 11/494,278, entitled “Patent Mapping,” by Steven W. Lundberg, Janal M. Kalis, and Pradeep Sinha, filed Jul. 27, 2006, which is incorporated herein by reference; and is further related to U.S. patent application Ser. No. 11/888,632, entitled “Patent Tracking,” by Steven W. Lundberg and Janal M. Kalis, filed Aug. 1, 2007 which is incorporated herein by reference; and is further related to U.S. patent application Ser. No. 10/710,656, entitled “Patent Mapping,” by Steven W. Lundberg, Janal M. Kalis, and Pradeep Sinha, filed Jul. 27, 2004 which is incorporated herein by reference and corresponding PCT application PCT/US2005/026768 filed Jul. 27, 2005.

BACKGROUND

[0004] Tools for identifying patents for a particular purpose such as a prior art search, validity analysis, or a freedom to operate investigation, operate by performing Boolean queries using various search operators. These operators allow for searching by date, terms, document number, and patent classification, among others. These tools further allow for searching individual document portions such as a document title, abstract, or claim set.

[0005] Other searching tools accept freeform text. Such tools accept a freeform text block and extract information from the text block deemed most likely to return acceptable results. However, such tools are still limited to only performing Boolean queries and displaying a list of results.

[0006] These search tools often provide large numbers of results, most of which are irrelevant. These tools fail to present results in a manner allowing for quick relevancy determinations. The presentation also fails to provide enough detail suggesting how to adjust a search for obtaining only relevant results. Further, the search tools provide the documents of the result set in a manner very similar to the traditional paper format of the documents. Quick relevancy determination is important in managing the assessment of large volumes of prior art for potential citation to government patent agencies.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Some example embodiments are illustrated by way of example and not limitation in the figures of the accompanying drawings in which:

[0008] FIG. 1 is a diagram of a system, according to an example embodiment.

[0009] FIG. 2 is a block diagram of a server device, according to an example embodiment.

[0010] FIGS. 3-8 are data model diagrams, according to example embodiments.

[0011] FIGS. 9-15 are user interfaces, according to example embodiments.

[0012] FIGS. 16-22 are example generated charts, according to example embodiments.

[0013] FIG. 23 is a computer system, according to an example embodiment.

[0014] FIGS. 24-30 are further user interfaces, according to example embodiments.

[0015] FIG. 31 is a block diagram showing a method of managing prior art, according to example embodiments.

[0016] FIG. 32 is a use case diagram, according to example embodiments.

DETAILED DESCRIPTION

[0017] The following detailed description of the present subject matter refers to the accompanying drawings which show, by way of illustration, specific aspects and embodiments in which the present subject matter may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the present subject matter. Other embodiments may be utilized and structural, logical, and electrical changes may be made without departing from the scope of the present subject matter. References to “an”, “one”, or “various” embodiments in this disclosure are not necessarily to the same embodiment, and such references contemplate more than one embodiment. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope is defined only by the appended claims, along with the full scope of legal equivalents to which such claims are entitled.

[0018] The subject matter herein provides systems, software, methods, and data structures for patent mapping, ranking and rating of patents, searching, and generating visual representations of the patents and patent portfolios to quickly analyze the patents for many reasons including, but not limited to, claim coverage and value. In an example embodiment, a patent portfolio may comprise one or more patents that may or may not be commonly owned or related. The collection of patent portfolios and patents may be stored in one or more databases. A patent may belong to more than one portfolio at the same time. In an example embodiment, the underlying patents and patent claims included in each patent portfolio may be categorized by patent concepts (sometimes referred herein as concepts) such as scope concepts (SC) and technology categories (TC).

[0019] In an example embodiment, technology categories are categories that claims relate to, but are not necessarily limited to. For example, a claim to a pulse generator may be put in the technology category “pacemaker”, but not be limited to a pacemaker per se—perhaps the claim merely says that the pulse generator generates a pulse of certain type that is useful for pacing, but pacing does not appear in the

claim. Hence, the claim relates to the technology category “pacemaker,” but it is not limited to being a pacemaker.

[0020] In an example embodiment, scope concepts are concepts that a claim is limited to. This is contrast to technology categories, where the claim may be mapped to a TC but it not necessarily limited to it. A scope concept may defined in a way to give the concept a context that a user can understand without necessarily having to look at the corresponding claim language. For example, if the scope concept is “method or apparatus for cardiac rhythm management”, and it is mapped to claim A, then claim A by definition is limited to this application, such that if a target device does not perform cardiac rhythm management, then it would not infringe claim A.

[0021] In an example embodiment, there are two types of scope concepts: 1) high level scope concepts that are like technical categories in the sense they are broad and general and apply to many claims in a portfolio; and 2) scope concepts that are specific to a limited number of claims—for example all claims in a patent may be limited to a very specific distinguishing feature, and this feature could be the basis for a scope concept.

[0022] In some example embodiments, high level scope concepts may be defined prior to mapping, and then assigned as applicable. For example, several scope concepts like: atrial pacing, ventricular pacing, defibrillation method or device, etc, may be defined. Then a mapping team may go through all claims in a portfolio and map these scope concepts to claims that are limited to these concepts. After the mapping is complete, an analysis may be done showing how many claims in the portfolio are limited to each of these scope concepts, and the claims may be presented for each SC. This may be useful is disqualify claims that are not of interest to a particular target (e.g., if an analysis is being done to find a claim that covers an alleged infringer). In some example embodiments, specific scope concepts are mapped patent by patent or by patent family. These may enable a person to create one or two scope concepts that can be mapped across all claims in given patent, a family of patents, or across a portion of a patent portfolio. In order to effectively formulate a scope concept that may be globally useful across a patent portfolio, it may be useful to be able to examine multiple patent claims at the same time even if they are not all in the same patent or patent family.

[0023] FIG. 1 illustrates an example system to implement the methods described herein. Shown is a user **102** and a user device **104**. The user device **104** may be, for example, a personal computer, mobile phone, or personal digital assistant. The user device **104** may be a computer system as described in FIG. 17. Users of the system may include specialized personnel trained to map patent claims as well as personnel trained to analyze the resulting claim map. The user device **102** may communicate with a server device **106** over a network **108** (e.g., the Internet) using a variety of communication means including, but not limited to, wired and wireless communication. The server device **106** may be a computer system as described in FIG. 17. In an example embodiment, the user **102** requests patent claims **110** from the server device **106** and transmits concept mappings **112** back to the server device **106** through the user device **104** via the network **108**. In various embodiments, one or more software applications are executed on the user device which facilitate the interactions and data transmissions between user **102**, user device **104**, and server device **106**. Other

information needed to complete the methods described herein may be transmitted between the user device **102** and server device **106** according to example embodiments.

[0024] FIG. 2 illustrates an example server device **200**. In an example embodiment, the server device includes one or more modules, databases, and engines. The various modules, databases, and engines may interact with each other and may take on the functionality of other modules, databases, and engines. Databases, according to an example embodiment, generally refer to sets of data stored in tables and may be implemented using a variety of database solutions including Oracle and MySQL. Engines, according to an example embodiment, generally refer to the generation of a product/image that is presented to a user (e.g., a webpage). Modules, according to an example embodiment, generally refer to functionality or features of the system that a user may invoke. For example, the mapping module may provide the necessary logic to create a mapping between a concept and a patent claim. According to an example embodiment, server device **200** includes an account database **202**, a mining module **204**, visualization engine **206**, a web server engine **208**, a ranking module **210**, a patent database **212**, a valuation module **214**, a tracking module **216**, a concept database **218**, a patent claim database **220**, a patent set database **222**, a mapping module **224**, and an ontology database **226**.

[0025] In various embodiments, the modules, engines, and databases are implemented in a combination of software and hardware. For example, a mapping module can be stored as set of instructions stored on a machine-readable medium. The instructions can be executed on a processor and cause a machine to perform operations related to mapping. Additionally, the visual presentation of data in not limited to engines and may be done by modules as well. Similarly, engines may contain underlying logic dictating on how each engine functions and interacts with the user, software, and hardware of the system. In various embodiments, the modules, engines, and databases are combined.

[0026] In an example embodiment, the account database **202** includes data pertaining to the different users of the system. In some embodiments, different levels of user are defined. For example, an administrator level allows the creation of an ontology (e.g., a collection of patent concepts and keywords) and mapping of patent claims while an analysis level user may only mine the map for patent claims. The web server engine **208** may present webpages to the user via the user device. The webpages may include forms and user interfaces for the user to interact with such that the user may manipulate the underlying data stored on the server device on one or more databases.

[0027] In an example embodiment, databases **212**, **218**, **220**, **222**, and **226** store the underlying data that the server device interacts with and modifies according to user input. The patent database **212** may include information related to all the patents, patent applications and patent portfolios stored in the system such as title, filing data, assignee, etc. As used in this specification, the term “patent matter” is intended to include within its scope and meaning any matter patent-related matter or case, for example a granted patent, a pending patent application, or an invention disclosure statement. The patent database **212** may include information related to prior art reference citations, or “references”. The prior art references may be associated with one or more of patents and/or patent applications stored in the patent data-

base **212**. The references may have been cited against or be otherwise potentially relevant to one or more of the patents or applications stored in the patent database **212**. In an example embodiment and method, the prior art references may require relevancy assessment prior to submission and/or citation to the USPTO or other international government patent agency imposing disclosure requirements on applicants. The prior art references may be collected, reviewed and cited accordingly in example embodiments and methods described in more detail further below. In this specification, references to mapping include references to prior art mapping. In example embodiments, two types of portfolio may be created and stored in the patent database **212**, an “Analysis” portfolio and a “Prior Art” portfolio.

[0028] The concept database **218** may store all the concepts that have been defined either by the user or automatically by the system. The patent claim database **220** may include information related to patent claims including which patent they belong to as well as concepts that have been mapped to the patent claims. The patent set database **222** may store information on sets of patents that have been defined by the user. In an example embodiment, a patent set may be defined by exclusion mining (e.g., the set of patents that have NOT been mapped to a certain concept). The ontology database **226** may store information on a user defined set of concepts.

[0029] The mapping module **224**, in an example embodiment, enables a user to map a concept to a patent claim. For example, the user may create and define a patent concept which is then stored in the concept database. The user may then send an indication, through the user device, that a patent claim in the patent claim database **220** should be mapped to the new concept. The indication may take the form of a type of user input such as clicking on an interface using an input device. The server device may then store this mapping in the patent claim database. For example, a relationship between the patent claim and concept may be stored in one or more of the databases. The mining module **204**, in an example embodiment, allows a user to search through the data stored in the databases to find patent claims of interest. For example, a user may wish to find all the patent claims related to a gear used in a bicycle. Rather than having the user define what the gear is, the user may indicate to the mining module what the gear is not, by indicating what concepts do not apply (exclusion mining). The mining module may search the entire universe of claims in the patent claim database, or a portion of the patent claim database, and retrieve the remaining patent claims (those claims that not have the concept) and present them to the user.

[0030] The visualization engine, in an example embodiment, generates reports and visual depictions of the data contained in a set of claims. For example, the visualization engine may generate a spreadsheet with the concepts in the concept database as rows and the patent claims as the columns. Color coding may be used to signify where a patent claim has been mapped to a concept. In some example embodiments, a user of the system may add additional data that influences the spreadsheet created. Some example embodiments include the generation of competitor landscape, freedom to operate, product coverage, validity, valuation, white space analysis, and white space claim generation spreadsheets. In various embodiments, other forms of coding are used such as shading and patterning.

[0031] The tracking module **216**, according to an example embodiment, maintains information related to a specific patent, group of patents, or concept. For example, the tracking module may store information related to a patent’s prosecution and litigation history such as office actions or claim amendments. Alerts (e.g., electronic mail) may be sent to a user indicating a change in a patent or patent application.

[0032] The ranking module **210** and valuation module **214**, according to example embodiments, enable the user to provide additional information related to patents, patent claims, and concepts that may be used to determine a course of action such as abandoning a patent or pursuing research in a specific field. For example, a user may indicate a specific concept as being key to her business. Further, a lawyer or other trained patent professional may provide a ranking for each patent included in her portfolio related to scope, design around protection, and detectability effort. The system may take this knowledge and through the visualization engine generate a chart that shows the highest ranked patents that also include her important concept.

Data Models

[0033] In an example embodiment, data models are defined to store the information related to the patents being analyzed. FIGS. 3-8 illustrate example data models that may be utilized. These may be defined in any suitable programming language such as C, C++, Java, Ruby, etc., that allows the manipulation of data models. In some embodiments, data models are referred to as classes and both terms will be used in the following descriptions. Further, an object may refer to a specific instance of a class or data model. As one skilled in the art will recognize, there may be more than one way to define the models and the relationships between the models. The illustrated models are to be taken only as one way of implementing the systems and methods described in this application. FIGS. 3-8, in some example embodiments, provide the lower level details of the information stored in databases **212**, **218**, **220**, **222** and **226**.

[0034] FIG. 3, according to an example embodiment, illustrates data models related generally to mappable data. Shown are models and relationships for a Patent **302**, Mappable Data **304**, Patent Specification **306**, Global Patent Ranking **308**, Mapping **310**, claim **312**, Concept **314**, Concept Type **316**, Patent Inclusion **318**, Patent Relation **320**, Ontology Relationships **322**, Ontology **324**, Mapping Status **326**, and Ontology Concept **328**. Each model may contain one or more elements that are defined either by the system or a user. Further, as illustrated, some models are related to each in other in a one to many relationship. For example, an Ontology object **324** may be related to many Ontology Concept objects **328**.

[0035] In an example embodiment, Patent model **302** includes types of information related to a patent including, but not limited to, whether or not it is an application, the number of claims, when it was filed, what organization it may belong to, the serial number, and its status. As can be seen, each piece of information may have an associated class such as a Boolean or string. In some cases, the type is actually another class (e.g., global ranking has a class of Global Ranking). Further shown are the elements of a data model that relate to another data model. For example, example Mapping Status **326**, Patent Relation **320**, Patent Inclusion **318**, and claim **312** models all include an element

of patent with a class of Patent. This relationship allows the system to examine a Claim class and determine the Patent in which the Claim is included.

[0036] In some embodiments, the Mapping 310 data model defines persistent objects that define the relationships between the a concept (e.g., technology categories and scope concept), a claim, and an ontology. As shown, there are many elements that a Mapping 320 class may include, such as, but not limited to, citations, notes, ontology, concept type and claim. Further, in an example embodiment, many Mapping objects may be related to one Ontology object and one Claim object. Thus, if one were to examine a Mapping object, there would be a relationship defining the ontology to which the object belongs to as well as the claim to which it has been assigned. In addition, there may be an integer signifying the type of concept to which the Mapping object belongs. As data model Concept Type 316 suggests each type of concept may be enumerated as well as be defined by an integer value. For example, the concept of scope concept may be given the value of '1.' Also, the "object" element illustrated has an associated class of Concept 314. Accordingly, the Mapping object may be linked to an example concept that has been defined as "two wheeled transportation." The mapping operation element may define the relationship between the cited claim and the concept. For example, a concept may be directly mapped to a claim. Other possibilities are discussed further with reference to FIG. 10. Accordingly, a Mapping object may contain the following information with regards to some of the displayed elements.

- [0037] claim: Claim A
- [0038] conceptType: 1
- [0039] object: Two wheeled transportation
- [0040] ontology: Bikes
- [0041] mapping operation: Directly Mapped

In an example embodiment, a Mapping object is created each time the system receives an indication a concept is to be mapped to a claim. In an embodiment, an indication may be stored that a concept is not mapped to a claim.

[0042] In an example embodiment, a Concept 314 object is created for every user defined concept as well as any concept the system may define automatically. Each Concept 314 may contain, but is not limited to elements of, conceptType, description, hidden, intelliMapAllowed, keywordLabel, name, organization, and underReview. As discussed above, a Concept 314 object may contain an enumeration of the ConceptType 316 object. For example, the conceptType element may have an example value of "scope concept." The description element may describe when a concept should be applied to a claim or other helpful information relating to the concept. The intelliMapAllowed Boolean may indicate whether the system may automatically apply the concept to other claims included in the system. For example, a concept type might be "keyword." These keywords may be verbatim phrases or individual words in the claim. Thus, a user may be able to safely have the system search other claims and find the same keyword and automatically create Mapping 310 objects for the keyword and found claims. The intelliMapAllowed may indicate whether the system should search automatically for these keywords.

[0043] In an example embodiment, the OntologyConcept 328 class only contains two elements, ontology and concept. An Ontology Concept 328 may be created to signify the relationship between a Concept 314 object and an Ontology

324 object. As shown, an Ontology 324 object may include many OntologyConcept objects. Also, as shown, a Concept 314 object may belong to many OntologyConcept 328 objects.

[0044] Also shown in FIG. 3, is the PatentInclusion 318 object. A PatentInclusion Object may include elements of inclusionType, patent, patentSet, ranking, reviewed, and ruleType. A PatentInclusion 318 object may be used to signify the relationship between a patent and a patentSet. This relationship is more fully explained with reference to FIG. 4.

[0045] FIG. 4, according to an example embodiment, illustrates data models related, generally, to mining mapped data. Shown are models and relationships for PatentOpinion 402, IncludedClaim 404, ScheduledIntelliMap 406, PatentSet 408, ConceptExclusion 410, PortfolioDomain 412, PatentSearch 414, PatentInclusion 416, LocalPatentRanking 418, TextInclusion 420, ConceptInclusion 422, and Concept 424 classes. In some embodiments, classes with the same name as in FIG. 3 are defined similarly. For example, Concept class 424 may contain the same elements as Concept class 314. However, as illustrated, additional functions are included that may operate on the class. For example, function "createCopy()" is illustrated in Concept class 424.

[0046] In an example embodiment, the PatentSet 408 class operates as the central class for mining. As illustrated, many of the other classes shown relate to the PatentSet 408 class. A PatentSet object may have many PatentInclusion 416, ConceptExclusion 410, and IncludedClaim 404 objects. Also, in an example embodiment, a PatentSet object may have many ConceptInclusion 422 objects related to it by virtue of the PatentSearch 414 class. Through user interfaces presented to a user and user input, a Patent Set may be defined. This may be done by a user adding claims manually or by a more sophisticated method involving a user defining which concepts to exclude or include. The various data models support an almost endless amount of customization for users of the system in the creation of patent sets.

[0047] In an example embodiment, the created patent sets may be saved for future use, as well as themselves becoming the basis for creating a new patent set. This may enable a user to efficiently search through any number of patents. The system may operate in such a manner that when a request is made to retrieve patents included in a patent set, the system responds by applying the relationships defined by the objects for that patent set. For example, the ConceptExclusion objects. This execution method may allow newly mapped patents to be included or excluded from the patent set with no additional input from a user. Thus, if a user wishes to find the intersection between a patent set related to vehicles and a patent set related to audio, the most current mapped patents available will be presented. As will be discussed in greater detail with respect to portfolio mapping, the ability to create patent sets and combine them may greatly speed up the process of finding common concepts across patents.

[0048] FIG. 5, according to an example embodiment, illustrates data models related, generally, to annuity data. Shown are models and relationships for Patent 502, Claim-MappedUser 504, AnnuityInformation 506, PatentRanking 508, ScoringCriteria 510, and Scorer objects 512. In an example embodiment to further enable a person to quickly analyze a large group of patent claims, patents may be given a rating. In an example embodiment, only the broadest independent claim in each patent is given a ranking, as the

broadest claim will often have the most value. The patent claims may be ranked according to multiple criteria, including, but not limited to scope, detectability, and the ability to design around the patent. In an example embodiment, the ranking information may be stored in a PatentRanking object and retrieved through the Scorer interface. Each criteria may be given a weighting depending on the client's needs. For example, a client may decide that scope is twice as important as the other two criteria. Therefore, the formula to rank the patents may be:

$$0.5(\text{scope})+0.25(\text{detectability})+0.25(\text{design around}) \\ = \text{rating.}$$

Once all of the patents have been ranked, the results may be presented to the user in a web browser, in the form of a chart, or using any other suitable display mechanism.

[0049] An AnnuityInformation object may include information related to annuities for an issued patent. Depending on the rating and annuity information of a patent, a user may automatically let patents go abandoned, a user may be alerted, or an annuity may automatically be paid. Other example embodiments will be obvious to one skilled in the art.

[0050] FIG. 6, according to an example embodiment, illustrates data models related generally to patent tracking. FIG. 7, according to an example embodiment, illustrates data models related generally to products and features. FIG. 8, according to an example embodiment, illustrates data models related generally to technology hierarchies.

Portfolio Mapping

[0051] FIG. 9 illustrates an example user interface that may be utilized to facilitate the methods described to map patent claims, according to an example embodiment. Displayed is the title **902** of a patent portfolio, controls are also illustrated that allow a user to edit to the portfolio, list the patents in the portfolio, “quick rank,” and generate a panoramic claim map. Also shown is the “Default Ontology” **904** being used. In an example embodiment, “Quick Rank” allows a user to map all the patent claims in a patent to concepts at the same time. An ontology, in an example embodiment, includes the different concepts available to a user to map to one or more of the patent claims. Further, there is an example search criteria box **906** which allows a user to specify a search query. Included are options to narrow the search by type of claim **908** including searching independent claims, dependent claims, or both. The search expression box **910** may allow a user to specify a regular expression to use as a search query. There is also an option to have keywords highlighted **912** in the search results. In an example embodiment, this may include the searched for keywords or keywords that have previously been mapped to the claims. Also shown are options to narrow the search results by technology categories **914** and scope concepts **916**.

[0052] FIG. 10 illustrates a method to map concepts to patent claims according to an example embodiment. A user interface such as the one illustrated in FIG. 9 may be used to facilitate this example method. Further, in an example embodiment, the method may be implemented using the data models and server device described above (e.g., server device **106** with reference to FIG. 1). At block **1002**, a database of patent portfolios and a database of patents are maintained, each patent stored in the database of patents

associated with one or more patent portfolios stored in the database of patent portfolios. A database management system may be used (DBMS) for storing and retrieving data from a data store which includes the database of patents and database of patent portfolios. In some embodiments, the DBMS is a relational database management system (RDBMS). In some other embodiments, the data store includes storing data in a Resource Description Framework Schema (RDFS). In some embodiments, communication with the data store includes using a language such as Structured Query Language (SQL) or extensible Markup Language (XML).

[0053] In an example embodiment, a database of ontologies may also be maintained, the ontologies including one or more patent concepts. As discussed above, an ontology may include all the metadata (patent concepts) that one may wish to map to a patent claim. For instance, the one or more patent concepts may include a technology category. The one or more patent claims may also include a scope concept, the scope concept defining a scope to which a patent claim is limited. The one or more patent or scope concepts in the database of ontologies may be organized in a hierarchical manner. Concepts lower in the hierarchy will typically be subsumed within the meaning or fall within the scope of a concept higher in the hierarchy. In an example embodiment, an indication of a relationship between a given patent concept and one or more patent claims in the database of patent claims may be stored. In further example embodiments, an indication stored against a concept higher in the hierarchy of patent concepts causes the same indication to be stored against associated concepts lower in the hierarchy of patent concepts. This ability to map and indicate claims against concepts collectively expedites and facilitates mapping review as several patent or scope concepts may be reviewed and dealt with at one time.

[0054] Keywords may also be used as patent concepts. These may be any term or short phrase that appears in the claim, exactly as it appears in the claim. As these terms are taken from the claims, they may be thought of as limitations in the sense that if the term cannot be read on an accused device, the claim probably does not cover the accused device. Example user interfaces showing scope concepts in an ontology can be seen with reference to FIG. 11.

[0055] In an example embodiment, at block **1004**, a search query associated with a first patent portfolio is retrieved. A user of the system may wish to search a previously created portfolio of patents. A patent portfolio may include patents that a user wishes to analyze. For example, a portfolio might include all of the patents for a company ABC Corp (ABC). A portfolio may be stored and defined as a patent set in the patent set database (e.g., patent set database **222** in FIG. 2) ABC might have received information on a potential infringing product. In order to find the patent claims relevant to the product, ABC may wish to map its entire patent portfolio and use the resulting mapped portfolio to quickly find the best claims to assert in an infringement lawsuit. However, it may also be useful to map the patents of the alleged infringer. These patents may also be added to the portfolio as it is likely ABC's patents and the alleged infringer's patents will have overlapping subject matter.

[0056] The search query may help to narrow down the patent. In an example embodiment, the search query may include a regular expression. For example, if the search query is “*” all the patent claims in the patent portfolio will

be displayed. Boolean expressions such as “car && dog” may also be used. In some example embodiments, an option is included to only search independent claims, dependent claims, or to search both. In some example embodiments, the portfolio may further be narrowed by using patent concepts that have been included in the current ontology.

[0057] FIG. 12 shows an example user interface with example options available to search by technology category. An example option is presented allowing a user to search technology categories disjunctively or conjunctively. In an example embodiment, each technology category in the ontology is shown to the user with three example options “Direct mapped claims,” “Direct Mapped or ‘Does Not Map,’” and “Direct Mapped or ‘Unresolved.’” These terms will be discussed in greater detail with respect to block 1010.

[0058] FIG. 13 shows an example user interface with example options available to search by scope concept. In an example embodiment, each scope concept in the ontology is shown to the user with four example options “Direct mapped claims,” “Do not include Direct Mapped or ‘Does Not Map,’” “Direct Mapped or ‘Does Not Map,’” and “Direct Mapped and ‘Unresolved.’” These terms will be discussed in greater detail with respect to block 1010.

[0059] Referring back to FIG. 10, in an example embodiment at block 1006, the first portfolio is searched as a function of the search query. At block 1008, in an example embodiment, search results 918 are generated, the search results including one or more patent claims associated with the search query. Using the search query provided, a query may be formatted as an SQL query or other suitable format to query the underlying databases. Generating the search results may include retrieving patent claims which include terms from the search query and synonyms of the terms as well as plural versions of terms in the search query. The results of the query may then be presented to the user in an example user interface as shown in FIG. 9. Only one patent claim is illustrated, however, more patent claims may have resulted from the search and may be shown simultaneous as to have the ability to manipulate multiple patent claims. Column headings may include four radio buttons 920 signifying the options available for mapping, the matter number, the claim number, the claim text, and other technology categories or scope concepts currently mapped to the claim. Because the generated search results are searching an entire portfolio of patents it may be possible that not all of the claims of a given patent will match to the search query. A trio of numbers 922 may also be displayed for each claim in relationship to the technology category heading and the scope concept heading. These represent the nature of the relationship between the claim and the technology category or scope concept. For example, as illustrated, claim 1 has two technology categories directly mapped: “space vehicle” and “Electric Device or Method.” It also has one scope concept directly mapped.

[0060] Referring back to FIG. 10, at block 1010, in an example embodiment, a plurality of patent claims are mapped to a patent concept. In an example embodiment, in addition to the database of patents and patent portfolios, a database of patent claims may be maintained. The database of patent claims may be administered and interacted with using a DBMS as described above. As described more fully with reference to FIG. 3, each patent claim may have one or more patent concepts that have been mapped to the claim. As discussed above with reference to narrowing down search

results, the relationship between a patent concept and a patent claim may take on many forms. For example, the relationship may be one where the patent concept is directly mapped to a patent claim. This may indicate that a user who looked at the claim made the decision that the patent claim was in a particular technology category, for example. Another relationship may indicate that a patent concept is not mapped to a patent claim. If a user is sure that a particular claim is not in a technology category, for example, it may be beneficial for that information to be saved so that the mapping process is not unnecessarily duplicative.

[0061] With reference back to FIG. 9, a user may select one or more patent claims to map based on the radio buttons displayed. In an example embodiment, there are four radio buttons indicating options for the claim: “Direct Mapped,” “Does Not Map,” “Unresolved,” and “No Operation.” The first two options are described in detail above. The “Unresolved” radio button may indicate that a user is not sure whether the concept should be mapped to the patent claim. This may be helpful in cases where the user does not have the legal or technical expertise to make a decision one way or another. A more senior user may then review the unresolved patent claims en masse at a later time. The last radio button may indicate that a user does not wish to have any relationship defined between the patent claim and a patent concept. In an example embodiment the “No Operation” radio button is selected by default for all the patent claims returned from the search query.

[0062] Upon a user indicating a preferred mapping for each patent claim, a user may further indicate a preference of which category of patent concept to map. In an example embodiment, there are two categories: technology categories and scope concepts (e.g., elements 924 and 926 in FIG. 9). In an example embodiment, a user clicks on the button corresponding to their preference and this preference is sent to the service device which detects the category of concept the user clicked. In response, a user interface is presented to the user corresponding to his or her preference. For example, FIG. 14 may be presented.

[0063] FIG. 14 illustrates an example search box 1402 and an example search results 1404 section. Across the top is an option to add a new patent concept 1406 (see FIG. 15 for a more detailed look at an example method to add a patent concept). Other options may include returning to the main mapping screen (e.g. FIG. 9) or canceling the mapping. In an example embodiment, the search box allows a user to search across an entire ontology for potential patent concepts. Similar to searching for patent claims, a user may enter a regular expression such as “*” to retrieve all the concepts included in the present ontology. For example, the results of the search 1404 displayed in FIG. 14 only returns “multiple blades.” A checkbox is presented allowing a user to select the concept 1408. If there is more than one concept displayed a user may select more than one of the concepts by selecting the respective checkboxes next to the patent concepts. A checkbox at the top of the results may be selected if a user wishes to select all 1410 of the concepts returned from the ontology search.

[0064] Further example options may be presented to the right of each concept. An option to “modify and map” 1412 may be selected if the user wishes to modify the concept. This may be useful if a user wishes to broaden the concept so that it may be mapped to more patent claims. For example, a narrow technology category may have been

defined as “power computer speakers.” Rather than a user defining a new technology category of “passive computer speakers,” the user might decide it makes more logical sense to only have one technology category titled “computer speakers.” In an example embodiment, the user can safely select “modify and map” and change the technology category to “computer speakers.” This may safely be done because all “powered computer speakers” are also “computer speakers.” In an example embodiment, every patent claim, regardless of which portfolio(s) it may be included in, will be updated to reflect the modified concept. Thus, it may not be advisable to narrow a concept without being certain every patent claim in the system adheres to the modified concept. In an example embodiment, a warning may be displayed to the user explaining the effects of modifying a concept.

[0065] In an example embodiment, an indication of a relationship between the patent concept and the plurality of claims in the database of patents may be stored (e.g., updating one or more databases). This may be accomplished, for example, by the user selecting the map button as displayed in FIG. 14. As discussed above, a user may have selected multiple patent claims resulting from searching and may have further indicated a preference for one or more patents claims to have concepts directly mapped as well as indicated a preference to have one or more patent claims have patent concepts not mapped. Thus, in an example embodiment, the indication of the relationship may include an indication that the plurality of patent claims are not mapped to the patent concept. In an example embodiment, the indication of the relationship includes an indication that the plurality of patent claims are mapped to the patent concept.

[0066] It some example embodiments mapping a plurality of patent claims to a patent concept includes defining the patent concept. Defining the patent concept may be initiated by a user clicking on the example “add_new” button 1406 as displayed in FIG. 14. An example user interface that may be presented to the user in response to this selection is illustrated in FIG. 15. Two example options may be presented, “Save” and “Cancel.” Also shows are two input text boxes, “Concept Name” and “Description.” The concept name may be the actual concept and may be, for example, either a technology category or a scope concept. A user may indicate which category of patent concept the new concept belongs using a pull down menu. The user may further wish to add the concept to an existing ontology by selecting one or more ontologies as presented in FIG. 15.

Report and Chart Generation

[0067] As described, the system may allow the generation of visual representation of the data included in the databases to further maximize the value of concepts to patent claims. In some example embodiments the charts may be interactive. In some example embodiments, a method to generate the charts includes formulating a query to send to one or more databases, the query requesting whether or not a set of patent claims have been mapped to a set of patent concepts. An additional query may be sent to the databases to determine additional metadata about the patent claims including, but not limited to, the filing data and owner of each patent. Ranking data may be received for each patent concept retrieved from the databases. In an example embodiment, the system generates a relationship between a ranking, a

patent concept, and a patent claim and displays the relationship to the user in the form of a chart. The ranking data may be stored in the database or may be received from a user. In some embodiments, ranking data may include integer values of disparate range (e.g., 1-10 or 1-100) alphabetical letters (e.g., a grading scale of A-F), or any other means to characterize a claim or concept.

[0068] In an example embodiment, a competitor landscape chart may be generated. An example simple competitor landscape chart is shown in FIG. 16. Shown is the title, patent number, filing data, total claims, each independent claim, and owner of each patent in a patent set. The patent set may reflect the patents owned by the competitors of a company requesting the map. The scope concepts that have been mapped to the patent claims in the patent set may be displayed as rows in the chart. If a scope concept has been mapped to a patent claim than the intersecting cell between the patent claim and concept may be filled in, checked, change color, patterned, shaded or otherwise have an indication of the mapping. Additional columns may indicate the competitor that first introduced a concept as well as the date it was introduced. This may be determined by examining each patent that has a concept mapped to at least one of the claims and examining the dates of each of the matching patents. In addition, the scope concepts may be sorted by frequency. Each competitor may be assigned a color or other designation such that a user may quickly determine which company owns each patent as well as who introduced what concepts first.

[0069] In an example embodiment, a product coverage chart may be generated. An example product coverage chart is shown in FIG. 17. As with FIG. 16, relevant patent information for patents included in the patent set is displayed. Also, displayed is a column titled “Has Feature?” This column may have values ranging from one to three signifying the degree to which the product includes the concept. For example, a value of three may mean the product definitely includes the concept and a value of one means the product definitely does not relate to the concept. Based on these values, the chart may dynamically update and determine values for the claim coverage and product coverage rows. For example, SC 1 and SC 5 both have a rating of ‘3’ and are present in all of the claims of patent “Title 1.” Thus, “Title 1” has a claim coverage of ‘3’ and a product coverage value of “potentially applies.” It is only “potentially applies” because it cannot be known for certain whether the patent applies but only that that SC 1 and SC 5 are present in the claims. However, because scope concepts always describe limitations, if a claim has two scope concepts mapped, as shown with respect to the claims in patent “Title 2,” the lowest ratings score will control the claim coverage. As shown, SC 6 is described as not being present in the product and the “Title 2” claims have been mapped to SC 6. Therefore, regardless of the fact the SC 3 has also been mapped and the product has this feature, the “Title 2” claims cannot apply to the product as they are at least limited to SC 6.

[0070] This chart may also allow interactivity with a user. This interactivity may include the user changing the “Has feature” values and the chart automatically updating the claim coverage and product coverage rows. For example, if the SC 1 rating was changed to ‘1,’ the claim coverage value of the patent “Title 1” may change to ‘1’ signifying the patent does not apply to the product. In some embodiments,

any changes that result from input from the user are highlighted on the chart. This may allow a user to quickly see the effects of potential changes to product coverage.

[0071] FIG. 18 illustrates an example freedom to operate chart. In an example embodiment, a freedom to operate chart allows a user to quickly see which patents may be necessary to obtain licenses from or purchase to produce a product, sell a service, etc. FIG. 18 is similar to FIG. 17 except for in place of a “Has feature?” column there is a “Need Feature?” column. The values in this column may represent whether or not, and to what extent, a user believes a feature is necessary in his or her product. A ‘3’ may indicate that the scope concept is necessary, a ‘2’ may indicate the scope concept is wanted, but not needed, and a ‘1’ may indicate the scope concept is not needed. In an example embodiment, the “claim status” row shown in FIG. 18 reflects whether a patent in the patent set needs to be licensed or purchased in order for the user to operate freely. As with the product coverage chart, the lowest value in the “need feature” column controls. Thus, patent “Title 1” is not needed even though SC 1 is mapped to all the claims and the user has indicated the feature is necessary. Similarly to above, a user may interact with the ratings to see in real-time the impact of removing or adding features in terms of the number of patents needing to be licensed or purchased.

[0072] FIG. 19 illustrates an example claim/patent valuation chart. In an example embodiment, a claim/patent valuation chart allows a user to see which patents/claims may be necessary to license, etc., to maximize the value of a currently owned patent or patent claim. FIG. 19 is similar to FIG. 17 except the values in the ratings column reflect whether or not a scope concept is necessary to maximize a patent’s value. The values in this column may represent whether or not, and to what extent, a user believes a feature is necessary in maximize a patent’s value. A ‘3’ may indicate that the scope concept is necessary, a ‘2’ may indicate the scope concept is wanted, but not needed, and a ‘1’ may indicate the scope concept is not needed. In an example embodiment, the “claim value” row shown in FIG. 19 reflects whether a patent in the patent set needs to be licensed or purchased in order to maximize the user’s patent. As with the product coverage chart, the lowest value in the rating column controls. Thus, patent “Title 1” is not important, even though SC 1 is mapped to all the claims and the user has indicated a high value for SC 1. Similarly to the above charts, a user may interact with the chart by changing the ratings to see in real-time the effects on patents in the patent set.

[0073] FIG. 20 illustrates an example validity chart. In an example embodiment, a validity chart allows a user to see the overlap between a patent and a patent set. FIG. 20 is similar to FIG. 17 except the values in the ratings column reflect whether or not a feature is shown in the patent in question (the patent to which the patent set is being compared). The values in this column may represent whether or not, and to what extent, a user believes a feature is present in the patent in question. A ‘3’ may indicate that the scope concept is shown, a ‘2’ may indicate the scope concept is possibly shown, and a ‘1’ may indicate the scope concept is not shown. In an example embodiment, the claim status row shown in FIG. 20 reflects the extent to which the patent in question and the patents in the patent set overlap. Unlike the product coverage chart, if two scope concepts are mapped to a patent, but contain different ratings, the feature rating

becomes “some overlap”. Thus, even though SC 5 has been rated as not shown and mapped to patent “Title 1,” SC 1 is also mapped to the patent but is shown and therefore there is some overlap between the patent in question and patent “Title 1.” A finding of “complete overlap” may indicate to a user that a patent or claim is completed anticipated. As above, the user may interact with the chart by changing the ratings to see in real-time the effects on patents in the patent set.

[0074] FIG. 21 illustrates an example white space analysis chart. In an example embodiment, a white space analysis chart allows a user to see the frequency in which scope concepts appear in a patent set. In some embodiments there are scope concepts that are in no patent claims. The chart may be color coded to allow a user to quickly ascertain the least frequently used scope concepts. FIG. 22 illustrates an example white space claim generation chart. In an example embodiment the generated chart illustrates suggested combinations of unclaimed combinations of existing scope concepts as well as suggested combinations of new scope concepts with existing scope concepts.

Computer System

[0075] FIG. 23 shows a diagrammatic representation of a machine in the example form of a computer system 2300 within which a set of instructions for causing the machine to perform any one or more of the methodologies discussed herein may be executed. In alternative embodiments, the machine operates as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine may operate in the capacity of a server or a client machine in a server-client network environment or as a peer machine in a peer-to-peer (or distributed) network environment. The machine may be a Personal Computer (PC), a tablet PC, a Set-Top Box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. Example embodiments can also be practiced in distributed system environments where local and remote computer systems which are linked (e.g., either by hardwired, wireless, or a combination of hardwired and wireless connections) through a network both perform tasks. In a distributed system environment, program modules may be located in both local and remote memory-storage devices (see below).

[0076] The example computer system 2300 includes a processor 2302 (e.g., a Central Processing Unit (CPU), a Graphics Processing Unit (GPU) or both), a main memory 2304 and a static memory 2306, which communicate with each other via a bus 2308. The computer system 2300 may further include a video display unit 2310 (e.g., a Liquid Crystal Display (LCD) or a Cathode Ray Tube (CRT)). The computer system 2300 may also include an alphanumeric input device 2312 (e.g., a keyboard), a User Interface (UI) cursor controller (e.g., a mouse), a disc drive unit 2316, a signal generation device 2318 (e.g., a speaker) and a network interface device (e.g., a transmitter) 2320.

[0077] The disc drive unit **2316** includes a machine-readable medium **2328** on which is stored one or more sets of instructions **2317** and data structures (e.g., software) embodying or utilized by any one or more of the methodologies or functions illustrated herein. The software may also reside, completely or at least partially, within the main memory **2304** and/or within the processor **2302** during execution thereof by the computer system **2300**, the main memory **2304** and the processor **2302** also constituting machine-readable media.

[0078] The instructions **2317** may further be transmitted or received over a network (e.g., the INTERNET) **2326** via the network interface device **2320** utilizing any one of a number of well-known transfer protocols (e.g., HTTP, Session Initiation Protocol (SIP)).

[0079] The term “machine-readable medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “machine-readable medium” shall also be taken to include any medium that is capable of storing, encoding or carrying a set of instructions for execution by the machine and that cause the machine to perform any of the one or more of the methodologies illustrated herein. The term “machine-readable medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic medium.

Prior Art Citation and Management

[0080] FIGS. **24** to **28** illustrate examples of user interfaces that may be utilized to facilitate the methods described to create and manage prior art portfolios and to map, assess and display an indication of the relevancy of prior art. In an example embodiment, patent claims may be mapped against one or more of the patents stored in a prior art patent portfolio. The prior art patents may be assessed for relevancy before potential submission to the USPTO or other international patent office according to any one or more of the patent mapping example embodiments described in this specification. In an example embodiment, two types of portfolio may be created and stored in the database **212** of patents and portfolios, an “Analysis” portfolio and a “Prior Art” portfolio. These portfolios may be used in many different ways, but an “Analysis” portfolio will typically be used for example in freedom-to-operate or “white space” investigations, while a “Prior Art” portfolio will be used for example in validity or relevancy assessments.

[0081] Example embodiments of the “Prior Art” portfolio may be used in the course of patent prosecution or in the preparation of Information Disclosure Statements. Prior art may be collected manually or automatically from one or more sources and reviewed for potential citation. Relevancy assessments may be displayed in one or more of the exemplary charts described in this specification. Prior Art portfolios may be created for the purpose of assembling related patent matters into groups so that prior art references cited, for example, against one of the matters within the group may propagate or “flow” into other matters for consideration as downstream or related citations in those other matters. This holistic overview and assessment of prior art greatly facilitates an applicant in discharging his or her duty to make full disclosure of all potentially relevant subject matter. A Prior Art portfolio will have all the features and capabilities of an

Analysis portfolio and may be used in mapping techniques as described more fully above under the heading Portfolio Mapping.

[0082] As shown in FIG. **24**, a visual indicator **240** is displayed in a Prior Art portfolio at graphic user interface (GUI) level in order to distinguish this portfolio. A portfolio “type” field **242** may also be introduced in similar way.

[0083] Cross citation of prior art references in portfolios may be accomplished using one or more of the functional modules described above in relation to FIG. **2**. References added to a patent matter listed in a selected Prior Art portfolio will automatically propagate or “flow” to other matters listed in that portfolio and into all other Prior Art portfolios where common matters exist between the portfolios until all the possibilities of reference flow (described further below) have been exhausted. At least one of the matters within a Prior Art portfolio must exist in another Prior Art portfolio in order for the added references to flow outside of that portfolio. This flow only occurs between Prior Art portfolios, and does not occur for example between a Prior Art portfolio and an Analysis portfolio. For example, a reference will not flow into any matters in Prior Art portfolio **P2** below if any reference is cited or added to a matter in Prior Art portfolio **P1**, since there are no common matters between these two portfolios. Similarly, no references will flow from either of Prior Art portfolios **P1** or **P2** to Analysis portfolio **PA1** since these portfolios are of different type, notwithstanding that matters **C** and **E** are respectively shared or “common”.

[0084] Prior Art **P1**—matters **A**, **B**, **C**

[0085] Analysis **PA1**—matters **C**, **D**, **E**

[0086] Prior Art **P2**—matters **E**, **F**, **G**

[0087] If an Analysis portfolio is converted into a Prior Art portfolio (by redesignation of the “type” field selector **242** described above, for example), the existing references cited against the various individual matters in this portfolio will flow to all other matters contained within the portfolio. In other words, all matters will have the same set of references for review, or mapping and/or potential submission to the USPTO, for example. Additionally on conversion of an Analysis portfolio to a Prior Art portfolio, references will flow to other Prior Art portfolios based on the existence of “common” matters as described above. In order to avoid a proliferation of potentially confusing reference flow, “reverse” or “roll back” reference flow is not allowed (i.e. is disabled) when a Prior Art portfolio is converted into an Analysis portfolio. All references associated with that portfolio will remain as they are at that point in time, but no further flow will occur if new references are added to the (converted) Analysis portfolio.

[0088] Certain further rules and exceptions may be provided for reference flow. A reference cannot flow into a matter having the status of one of the following types: Issued, Archived Expired, Abandoned, Expired, Expired Pending Renewal/Extension, and Expired/Terminated. New references flowing into matters are initially referenced as “Unmarked”. Duplicate references cannot flow into any matter. For example, if a matter already contains Version 1 of reference **R**, the system will not allow Version 2 of Reference **R** to flow into the same matter.

[0089] In some example embodiments, the propagation or “flow level” of references flowing between respective Prior Art portfolios may be indicated. The term “flow level” can also be referred to as “generation level” or “generation

flow". Such an indication can serve to categorize the prior art citations and may greatly assist in determining the likely relevance of a listed prior art reference to be cited in a potential Information Disclosure Statement, for example. A "source" patent matter, in a given patent portfolio, that a prior art reference is added to or associated with initially does not have a "flow level" or "level indicator" since that matter is indicated as the "source" matter for that reference. This "source" level may be considered as a "zero" or "base" level for the reference. FIG. 25 indicates such a designation at 250.

[0090] All matters in the portfolio in which that "source" matter is contained receiving the "flow" of that reference to them will exhibit a "Flow Level" indication of "1". When the reference flows outside the portfolio, for example, where a "common" matter exists between portfolios, a "Flow Level" indication of "2" is displayed. Further "flow" of that reference into other portfolios will yield "Flow Level" indications of "3", and so on. The further a reference "flows" from its source, the higher the flow level indication. Potential relevancy of a given reference may be considered inversely proportional to its flow level.

[0091] In example embodiments, "Status" indicators may be provided for prior art references in the course of their initial identification and subsequent flow between portfolios. In FIG. 26 at 260, a status indication of "NUL" indicates that a reference is not applicable as or does not constitute prior art. For example, the referenced document may simply be a formal notice from a patent office indicating that an action is outstanding. The notice itself does not contain any technical disclosure pertaining to the relevant subject matter. "UNMARKED" is the default status indication provided when a new reference flows to a patent matter under reference flow. "CITED" is designated by a user when a reference is submitted to a patent office as part of an Information Disclosure Statement, for example. "RECEIVED FROM PTO" is designated when a reference is received from the relevant Patent Office, typically after examination or search. "DISMISSED" is designated by a user when a reference is considered not to pertain to a matter i.e. it is not relevant subject matter for submission under an Information Disclosure Statement, for example. In this instance, the mapping or other modules (see FIG. 2) operate to "hide" this reference from further consideration in the matter it has been dismissed from, but the reference is still available for "flow" to other matters within the portfolio or to other portfolios. In some example embodiments, a check box (for example shown at 270 in FIG. 27) is provided which can be selected to display all dismissed references in a listing. A user can change the reference status at any time and may do so collectively, or "in bulk", in an example embodiment. The "bulk" selection can be used by check box 272 in FIG. 27, for example. A user can select multiple references from a reference list and can change the status of all selected references at one time. There is no restriction on what a status can be changed to or from. A mismatched reference can be corrected, for example.

[0092] In other example embodiments and with reference to FIG. 28, a source date column is added to the user interface at 280. The source date is stored in the patent database and is displayed in the reference listing screen to indicate the date the reference was first introduced into or associated with the source matter. This source date may be

useful in assessing relevancy when the reference flows to other matters or other portfolios.

[0093] Displayed at 290 in FIG. 29 is a "References" tab which may be selected to generate a list of references contained in a given portfolio against one or more matters listed in that portfolio. In some example embodiments shown in FIG. 30, sub-tabs 312 with associated functionality are provided. The sub-tabs may include sub-tab 300 (All References), sub-tab 302 (Patent References), sub-tab 304 (Non-Patent References), sub-tab 306 (Bulk Add), sub-tab 308 (Job Status) and sub-tab 310 (Holding List). This additional functionality allows, for example, a user to add (or "scrape") references into a given matter or portfolio "in bulk", or selectively display references under one or more different headings or reference types (patent, non-patent etc).

[0094] FIG. 31 illustrates a method to map concepts to patent claims according to an example method embodiment. A user interface such as any of those illustrated in FIGS. 24 to 30 may be used to facilitate this example method. Further, in an example embodiment, the method may be implemented using the data models and server device described above (e.g., server device 106 with reference to FIG. 1), or using any of the modules described with reference to FIG. 2. At block 3102, a database of prior art portfolios, a database of patent matters and a database of prior art reference citations (or "references") are maintained, at least some of the prior art reference citations being associated with one or more of the patent matters stored in the database of patent matters. At block 3104, a selection or sub-set of patent matters is loaded into a first prior art portfolio for further analysis and potential cross-citation to other patent matters and/or prior art portfolios. At block 3106, at least one initial or "source" patent matter contained in the first prior art portfolio is identified. At block 3108, any one or more of the prior art reference citations cited against (or otherwise associated with the source patent matter by reason of potential subject matter relevancy, for example) is caused to flow and thereby be associated with at least one other patent matter in the prior art portfolio. In this way, the references cited or associated with a "source" matter in a given prior art portfolio can be cross-cited for further study (and potential citation to a national patent office) in another "downstream" matter in the same portfolio or, in other example embodiments, a different portfolio.

[0095] At block 3110 in other example embodiments, the prior art references associated with the at least one source patent matter may be caused to flow and be associated with all the other patent matters in the same (first) prior art portfolio. At block 3112 in other example embodiments, the references may be caused to flow to at least one patent matter contained in a different (or second) patent portfolio.

[0096] At block 3114, in further example embodiments, the method may include providing an indication of the flow level of the prior art reference citations flowing from the at least one patent matter to the at least one other patent matter. As described above, this flow level indication may be designated as a "0", "1" or a "2" depending on the successive levels of flow of a given reference from matter to matter, or in other example embodiments from portfolio to portfolio. At block 3116, in further example embodiments, the method may include providing an indication of the date on which a reference is first associated with or cited against a source patent matter, or the date on which a reference flows

from the source patent matter to another matter or portfolio downstream. At block 3118, in further example embodiments the method may include providing a status indicator for each of the prior art references associated with the patent matters contained in at least the first prior art portfolio. At block 3120, in further example embodiments the method may further include providing an indication of the ground of rejection on which a prior art citation was cited by a national patent office to reject the claims of any one or more of the patent matters contained in at least the first prior art portfolio.

[0097] In further example embodiments, and with reference again to FIG. 2, the mapping module 224 may be configured to cause prior art reference citations associated with at least one source patent matter contained in a first Prior Art portfolio to flow to and be associated with at least one other patent matter contained in that Prior Art portfolio. The mapping module 224 may further be configured to allow references to flow to and be associated with patent matters in second or further Prior Art portfolios. The mapping module 224 may further be configured to identify and provide the flow level indications (“0”, “1”, or “2” etc), the date on which references flow from a source patent matter to another one, and/or the ground of rejection on which one or more of the prior art citations may have been advanced by a national patent office.

[0098] Referring again to FIG. 2, in further example embodiments, the web server engine 208 is configured to load a selection of patent matters into the first Prior Art portfolio. The patent matters may be selected manually by a user based on predetermined criteria or relevance to a target subject matter, or may be loaded automatically by the engine pursuant to a search conducted by the mining module 204 in response to a search query. The tracking module 216 is configured to identify any reference citations associated with a selected patent matter in the portfolio, referred to as “source” patent matter, in the Prior art portfolio. The tracking 216 or mapping module 224 is configured to associate the identified reference citations with at least one other patent matter in that Prior Art portfolio, or a second Prior Art portfolio. For clarity of description, this association process is called “flow” in this specification. The flow may be configured to occur automatically, or in response to a series of selections entered manually by a user.

[0099] A user may employ the methods, embodiments and systems of the invention to load matters (granted patents, pending applications, invention disclosures and the like) into identified “Prior Art” portfolios. Prior art citations associated with those patent matters can be caused automatically (or manually pursuant to specified criteria) to flow between matters in a portfolio, or between common matters listed in more than one Prior Art portfolio. The citations can be kept and viewed in lists in the portfolios, and the patent matters and associated citations can be added or deleted from portfolios pursuant to identified relationships (e.g. common priority claim, or same title etc), or not as the case may be. Any arbitrary patent matters or prior art reference citations can be added.

[0100] Prior art citations may be displayed in each matter and filtered for example by their nature or source (e.g. citations arriving from other matters or portfolios), the type of rejection (e.g. novelty 102/obviousness 103 etc), or by the number of flow levels (or “hops”) it took for a citation to

flow to and reach a given destination matter or portfolio from a source. Other forms of analysis may be employed to identify filtering criteria.

[0101] An example use case and further example embodiments are now described with reference to FIG. 32. Suppose Prior Art Portfolio 1 contains matters A, B and C while Prior Art Portfolio 2 contains matters C, D and E. Typically but not necessarily, matters A, B and C will be related to each other so as to be in the same patent family with, for example, a common priority claim under an international patent convention. The matters in Portfolio 2 will typically, but not necessarily, be related in some way to the matters in Portfolio 1. For example, matters D and E may be divisional cases of matter C. The typical situation is such that there is a duty to disclose and cross-cite potentially relevant prior associated with each of the related matters A through E.

[0102] In an example embodiment, Reference P1 is entered into Matter A either automatically using a “scraper” tool, for example, or manually pursuant to one or more selection criteria adopted by a user. At this stage, and in this example embodiment, the Reference P1 has a flow level of “0”. This level indicates that a reference was sent to a prior art portfolio for citation, but was not yet added to another individual matter (Flow level 1) within the portfolio before beginning to flow across matters and portfolios. In this way, a user is able to review references at “0” level before starting cross-citation or further propagation to other matters or portfolios. In further example embodiments, references at “0” level may be held in a “pre-portfolio” holding area without yet designating a single matter in that portfolio that would start the flow process. The holding area may be visually represented separately in any one of the user interfaces described above.

[0103] Referring back to FIG. 32, after entering Matter A, Reference P1 automatically flows into Matters B and C as P1-1 indicating flow level “1” or first generation flow. Reference P1 also automatically flows into Matters D and E in Portfolio 2 from common Matter C as P1-2, indicating flow level “2” or second generation flow. Further propagation or flow continues as long as the matters that Reference P1 has flowed into exist in other portfolios (with their proper generations indicated) until all possibilities are exhausted.

[0104] In some embodiments, a reference (such as P1) will not flow to be added into matters where it already exists (i.e. with which it is already associated), but the prior association in that matter of a flowing reference will not stop the further propagation of the reference. Hence, if Reference P1 in FIG. 32 was already associated with Matter D as P1-1 it would stay indicated as such but would still flow into Matter E as P1-2. In some embodiments, all references that flow automatically will flow into the related matters as “Unmarked” for their status indicating that they have not yet been marked in any fashion with regards to citing or not citing to a given local or international patent office.

[0105] The Abstract of the Disclosure is provided to comply with 37 C.F.R. § 1.72 (b), requiring an abstract that will allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed

embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

What is claimed is:

1. A system comprising:
at least one processor; and
memory comprising instructions that, when executed by the at least one processor, cause the at least one processor to perform operations to:
configure a computer system with specialized data processing architecture that improves patent analysis efficiency by deploying interconnected data storage systems for patent portfolios and patents, each patent computationally linked to one or more patent portfolios through relational data structures;
implement a technical mapping framework that creates computational relationships between patent claims and hierarchically organized patent concepts comprising both technology categories and scope concepts, wherein the technology categories comprise relational data indicating subject matter associations and the scope concepts comprise limitation data that defines claim boundaries for automated analysis, and wherein the technical mapping framework reduces search processing time by enabling concept-based queries rather than text-based searches;
receive a search query for target subject matter and processing the search query through the technical mapping framework to generate computationally filtered search results including one or more patent claims containing query terms and computationally determined synonyms;
automatically create persistent data associations between the one or more patent claims and at least one patent concept, wherein the persistent data associations are stored in specialized mapping data structures that enable rapid retrieval operations and include mapping objects with computational identifiers that link patent claims, patent concepts, and ontologies through relational database operations;
execute automated visualization algorithms that process the persistent data associations to generate interactive visual representations displaying claim-concept relationships, wherein the automated visualization algorithms computationally generate at least one of: competitor landscape charts, product coverage charts, freedom to operate charts, claim valuation charts, validity charts, and white space analysis charts; and
output the interactive visual representations to a display interface configured to enable real-time modification of the persistent data associations, wherein modifications trigger automatic recalculation and redisplay of visual representations to improve patent analysis workflow efficiency.
2. The system of claim 1, wherein the hierarchically organized patent concepts include computationally distinct technology categories and scope concepts, the computationally distinct technology categories comprising relational data indicating subject matter associations, and the scope

concepts comprising limitation data that defines claim boundaries for automated analysis.

3. The system of claim 2, wherein hierarchical organization of the hierarchically organized patent concepts includes parent-child data relationships that enable automated inheritance processing, wherein creating an association with a parent concept automatically triggers computational propagation to child concepts through the specialized mapping data structures.

4. The system of claim 1, the memory further comprising instructions that, when executed by the at least one processor, cause the at least one processor to perform operations to maintain ontology data structures that organize the hierarchically organized patent concepts, and wherein creating the persistent data associations includes generating mapping objects with computational identifiers that link patent claims, patent concepts, and ontologies through relational database operations.

5. The system of claim 1, the memory further comprising instructions that, when executed by the at least one processor, cause the at least one processor to perform operations to execute exclusion-based mining algorithms that process concept absence data to computationally filter patent claims and generate refined result sets, wherein a patent set is defined by exclusion mining comprising patents that have not been mapped to a certain concept.

6. The system of claim 1, the instructions to execute the automated visualization algorithms further comprising instructions to computationally generate at least one of: competitor landscape charts, product coverage charts, freedom to operate charts, claim valuation charts, validity charts, and white space analysis charts, each chart type requiring distinct algorithmic processing of the claim-concept relationships.

7. The system of claim 1, the instructions to process the search query further comprising instructions that, when executed by the at least one processor, cause the at least one processor to perform operations to execute semantic analysis algorithms that retrieve patent claims containing query terms and computationally determined synonyms, improving search accuracy over conventional text matching.

8. At least one non-transitory machine-readable medium comprising instructions that, when executed by at least one processor, cause the at least one processor to perform operations to:

configure a computer system with specialized data processing architecture that improves patent analysis efficiency by deploying interconnected data storage systems for patent portfolios and patents, each patent computationally linked to one or more patent portfolios through relational data structures;

implement a technical mapping framework that creates computational relationships between patent claims and hierarchically organized patent concepts comprising both technology categories and scope concepts, wherein the technology categories comprise relational data indicating subject matter associations and the scope concepts comprise limitation data that defines claim boundaries for automated analysis, and wherein the technical mapping framework reduces search processing time by enabling concept-based queries rather than text-based searches;

receive a search query for target subject matter and processing the search query through the technical map-

ping framework to generate computationally filtered search results including one or more patent claims containing query terms and computationally determined synonyms;

automatically create persistent data associations between the one or more patent claims and at least one patent concept, wherein the persistent data associations are stored in specialized mapping data structures that enable rapid retrieval operations and include mapping objects with computational identifiers that link patent claims, patent concepts, and ontologies through relational database operations;

execute automated visualization algorithms that process the persistent data associations to generate interactive visual representations displaying claim-concept relationships, wherein the automated visualization algorithms computationally generate at least one of: competitor landscape charts, product coverage charts, freedom to operate charts, claim valuation charts, validity charts, and white space analysis charts; and

output the interactive visual representations to a display interface configured to enable real-time modification of the persistent data associations, wherein modifications trigger automatic recalculation and redisplay of visual representations to improve patent analysis workflow efficiency.

9. The at least one non-transitory machine-readable medium of claim 8, wherein the hierarchically organized patent concepts include computationally distinct technology categories and scope concepts, the computationally distinct technology categories comprising relational data indicating subject matter associations, and the scope concepts comprising limitation data that defines claim boundaries for automated analysis.

10. The at least one non-transitory machine-readable medium of claim 9, wherein hierarchical organization of the hierarchically organized patent concepts includes parent-child data relationships that enable automated inheritance processing, wherein creating an association with a parent concept automatically triggers computational propagation to child concepts through the specialized mapping data structures.

11. The at least one non-transitory machine-readable medium of claim 8, further comprising instructions that, when executed by the at least one processor, cause the at least one processor to perform operations to maintain ontology data structures that organize the hierarchically organized patent concepts, and wherein creating the persistent data associations includes generating mapping objects with computational identifiers that link patent claims, patent concepts, and ontologies through relational database operations.

12. The at least one non-transitory machine-readable medium of claim 8, further comprising instructions that, when executed by the at least one processor, cause the at least one processor to perform operations to execute exclusion-based mining algorithms that process concept absence data to computationally filter patent claims and generate refined result sets, wherein a patent set is defined by exclusion mining comprising patents that have not been mapped to a certain concept.

13. The at least one non-transitory machine-readable medium of claim 8, the instructions to execute the automated visualization algorithms further comprising instructions to

computationally generate at least one of: competitor landscape charts, product coverage charts, freedom to operate charts, claim valuation charts, validity charts, and white space analysis charts, each chart type requiring distinct algorithmic processing of the claim-concept relationships.

14. The at least one non-transitory machine-readable medium of claim 8, the instructions to process the search query further comprising instructions that, when executed by the at least one processor, cause the at least one processor to perform operations to execute semantic analysis algorithms that retrieve patent claims containing query terms and computationally determined synonyms, improving search accuracy over conventional text matching.

15. A computer implemented method comprising:

configuring a computer system with specialized data processing architecture that improves patent analysis efficiency by deploying interconnected data storage systems for patent portfolios and patents, each patent computationally linked to one or more patent portfolios through relational data structures;

implementing a technical mapping framework that creates computational relationships between patent claims and hierarchically organized patent concepts comprising both technology categories and scope concepts, wherein the technology categories comprise relational data indicating subject matter associations and the scope concepts comprise limitation data that defines claim boundaries for automated analysis, and wherein the technical mapping framework reduces search processing time by enabling concept-based queries rather than text-based searches;

receiving a search query for target subject matter and processing the search query through the technical mapping framework to generate computationally filtered search results including one or more patent claims containing query terms and computationally determined synonyms;

automatically creating persistent data associations between the one or more patent claims and at least one patent concept, wherein the persistent data associations are stored in specialized mapping data structures that enable rapid retrieval operations and include mapping objects with computational identifiers that link patent claims, patent concepts, and ontologies through relational database operations;

executing automated visualization algorithms that process the persistent data associations to generate interactive visual representations displaying claim-concept relationships, wherein the automated visualization algorithms computationally generate at least one of: competitor landscape charts, product coverage charts, freedom to operate charts, claim valuation charts, validity charts, and white space analysis charts; and

outputting the interactive visual representations to a display interface configured to enable real-time modification of the persistent data associations, wherein modifications trigger automatic recalculation and redisplay of visual representations to improve patent analysis workflow efficiency.

16. The method of claim 15, wherein the hierarchically organized patent concepts include computationally distinct technology categories and scope concepts, the computationally distinct technology categories comprising relational data indicating subject matter associations, and the scope

concepts comprising limitation data that defines claim boundaries for automated analysis.

17. The method of claim 16, wherein hierarchical organization of the hierarchically organized patent concepts includes parent-child data relationships that enable automated inheritance processing, wherein creating an association with a parent concept automatically triggers computational propagation to child concepts through the specialized mapping data structures.

18. The method of claim 15, further comprising maintaining ontology data structures that organize the hierarchically organized patent concepts, and wherein creating the persistent data associations includes generating mapping objects with computational identifiers that link patent claims, patent concepts, and ontologies through relational database operations.

19. The method of claim 15, further comprising executing exclusion-based mining algorithms that process concept absence data to computationally filter patent claims and generate refined result sets, wherein a patent set is defined by exclusion mining comprising patents that have not been mapped to a certain concept.

20. The method of claim 15, wherein executing the automated visualization algorithms includes computational generation of at least one of: competitor landscape charts, product coverage charts, freedom to operate charts, claim valuation charts, validity charts, and white space analysis charts, each chart type requiring distinct algorithmic processing of the claim-concept relationships.

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